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BACHELOR OF SCIENCE IN MARINE ENGINEERING

**DEVELOPMENT AND EVALUATION OF AN AUDITABLE,
RESOURCE-CONSCIOUS PARAMETRIC OPTIMIZATION
WORKFLOW FOR MARINE STRUCTURAL COMPONENTS**

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by

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*A project report submitted in partial fulfilment of the requirements
for the award of the Degree of Bachelor of Science in Marine Engineering
at the Jomo Kenyatta University of Agriculture and Technology*

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DECLARATION

We hereby declare that this Project Report is our original work and has not been presented for a degree in any other university or for any other award.

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Supervisor Declaration

This Project Report has been submitted for examination with my approval as the University Supervisor.

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Dedication

We dedicate this Project Report to our parents, guardians, lecturers, and mentors. Their encouragement, sacrifice, and guidance supported our academic journey and the completion of this project.

Acknowledgements

We thank our supervisor for the technical guidance, critical questions, and academic direction provided throughout this project. We also thank the staff of the Department of Marine Engineering and Maritime Operations for the knowledge and facilities that supported this work. We are grateful to our classmates, friends, and families for their practical help and encouragement. Autodesk educational access and the open-source Python community made it possible to complete the software work without direct project expenditure.

Abstract

This project developed and evaluated the Nelder-Mead Marine Optimizer (NEMO), an auditable workflow for the parameterized mass optimization of marine structural components. The system combines unit-aware engineering definitions, first-order structural screening, seeded Latin-hypercube sampling, bounded multi-start Nelder-Mead search, traceable run records, a file-based Autodesk Fusion bridge, native CAD generation, semantic boundary metadata, validation-package generation, and a results dashboard. Three component archetypes were studied: an Aluminum 6061-T6 equipment bracket, an S275 steel lifting padeye, and an Aluminum 6061-T6 stabilizer fin. Each part used 60 seeded Latin-hypercube samples and three local-search starts. The lightest selected analytically feasible finalists had masses of 0.1134 kg, 3.5478 kg, and 14.3154 kg, which corresponded to reductions of 72.9%, 59.9%, and 27.2% from the respective analytical baselines. Native Fusion CAD mass differed from the analytical estimate by 1.15%, 3.80%, and 0.59% for these finalists. Generator sweeps tested each baseline and 20 seeded design vectors. The bracket and padeye completed all 21 integration cases. The stabilizer generated all 21 artifacts, although the outer test process reached its 30-minute timeout while reading the final response. The bracket also preserved the original proof-of-concept path into a manual Fusion Static Stress study. Inspection of the exported report found that the saved study used 1.472 N instead of the intended integer load of 1472 N. The FEA numerical results were therefore rejected, while the report images were retained as setup and quality-control evidence. The work shows that a resource-conscious, human-reviewed workflow can reduce routine search and CAD effort while keeping analytical, CAD, and FEA claims separate. The reported designs remain screening candidates. They require corrected, mesh-converged FEA, design-code checks, verified loads and materials, fatigue and fabrication assessment, and qualified engineering review before manufacture or service.

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LIST OF ABBREVIATIONS AND SYMBOLS

Abbreviations

API	Application Programming Interface
CAD	Computer-Aided Design
CAE	Computer-Aided Engineering
CFD	Computational Fluid Dynamics
CLI	Command-Line Interface
FEA	Finite Element Analysis
FOS	Factor of Safety
IPC	Inter-Process Communication
JSON	JavaScript Object Notation
JKUAT	Jomo Kenyatta University of Agriculture and Technology
LHS	Latin-Hypercube Sampling
NACA	National Advisory Committee for Aeronautics
NEMO	Nelder-Mead Marine Optimizer
POC	Proof of Concept
SI	International System of Units
STEP	Standard for the Exchange of Product Model Data
STL	Stereolithography mesh format
V&V	Verification and Validation

Symbols

A	Area (m^2)
b	Stabilizer span (m)
C_L	Lift coefficient
E	Young's modulus (Pa)
F_d	Design load (N)
$F_{\min}$	Minimum permitted factor of safety
g	Acceleration due to gravity (m/s^2)
I	Second moment of area (m^4)
$J(\mathbf{x})$	Penalized optimization objective
K_t	Stress-concentration factor
m	Mass (kg)
M	Bending moment (N m)
ρ	Density (kg/m^3)
σ	Normal or equivalent stress (MPa)

σ_y	Material yield strength (MPa)
τ	Shear stress (MPa)
V	Volume (m ³) or speed (m/s), as identified in context
w	Penalty weight
$\mathbf{x}$	Physical design vector
$\mathbf{z}$	Scaled design vector
δ	Displacement (mm)
$\delta_{\max}$	Maximum permitted displacement (mm)

Chapter 1

Introduction

1.1 Chapter Overview

This chapter introduces the engineering problem, the project aim, the research questions, and the boundaries of the reported evidence. It also defines the project contribution and explains why the three selected components provide a useful test of one shared workflow. The chapter closes with the structure of the report.

1.2 Background

Marine vessels and offshore systems contain many secondary structural components. Examples include machinery foundations, equipment brackets, lifting points, appendage roots, stiffeners, local doublers, and attachment details. A single component may be small compared with a hull or an offshore platform, but it can transfer a concentrated load into a much larger structure. Its behavior can therefore affect equipment alignment, lifting safety, watertight integrity, vibration, and the durability of adjacent structure.

The load path through a small component is often three-dimensional. A bracket can combine baseplate bending, rib bending, fastener reactions, bearing, contact, and local stress at a fillet or weld. A padeye can combine pin bearing, net-section tension, shear-out, out-of-plane bending, lug-root stress, gusset action, and weld load transfer. A stabilizer fin can combine distributed pressure, root bending, shear, torsion, shell action, spar behavior, local buckling, fatigue, and hydrodynamic effects. A credible design process must identify which of these effects the selected model can represent and which effects remain outside its intended use.

Mass is also an important design quantity. Added structural mass increases material use and can affect vessel displacement, payload, installation effort, fuel demand, and center-of-gravity management. However, mass reduction is not a sufficient design objective on its own. Removing material can reduce strength, stiffness, stability, fatigue life, corrosion reserve, fabrication tolerance, and damage tolerance. Marine component design

is therefore a constrained task: mass can be reduced only while applicable strength, displacement, stability, fatigue, manufacturing, inspection, and service requirements remain satisfied.

Traditional component development often moves between preliminary calculations, manual CAD edits, and selected FEA studies. This process can produce an acceptable design, but it normally examines only a small part of a multi-variable design space. The reasoning can also become fragmented. Dimensions may be held in a drawing, loads in a calculation sheet, materials in a simulation setup, boundary conditions in screenshots, and candidate history in personal notes. When the evidence becomes separated, a reviewer cannot easily reproduce the result or determine whether one image belongs to the same design as one table of values.

Parametric modeling provides a different basis for iteration. A parametric model represents a family of related designs rather than one fixed geometry. An optimization process can change the dimensions, evaluate each candidate, record its performance, and select candidates for more detailed work. Established research has connected commercial CAD and CAE tools into automated optimization loops [1]. This integration can reduce repetitive work, but it introduces a difficult problem: a geometry change can also change the topology used for loads, supports, contacts, and mesh controls. The model must preserve the engineering meaning of these regions, not only rebuild a visually plausible solid. Nolan *et al.* describe this need as the preservation of simulation intent [2].

1.3 Project Context

The Nelder-Mead Marine Optimizer (NEMO) was developed in response to the combined problems of design-space exploration, traceability, and CAD-analysis handoff. The external Python application stores part definitions, evaluates analytical models, samples the design space, runs the optimizer, records results, and creates validation packages. The Autodesk Fusion add-in rebuilds native CAD, calculates physical properties, exports neutral geometry, and describes the faces associated with engineering boundary roles.

The original proof-of-concept plan targeted a fully automatic cycle in which the optimizer would request a design, Fusion would rebuild the CAD and solve a Static Stress study, and the structural results would return to the optimizer. An implementation investigation found that the public Fusion Python API used by this project did not provide a dependable route to start the required Static Stress solves and extract all required result quantities. The project therefore changed to a tiered workflow. Low-cost analytical models guide the search, Fusion produces native CAD and mass evidence for selected designs, and an engineer prepares and checks final FEA studies manually.

This change reduced the degree of automation, but it created a clearer evidence boundary. A CAD-only response contains volume, mass, and artifact paths. It leaves stress, factor of safety, and displacement empty. The reserved open-source FEA mode

also returns a controlled failure because no Gmsh and CalculiX implementation currently exists. The software does not invent values when an evaluator is unavailable.

1.4 Problem Statement

This project addressed two related problems.

The first problem concerned engineering efficiency. Manual iteration does not systematically explore the combined effects of plate size, stiffener dimensions, fillet radius, lug geometry, spar position, and reinforcement dimensions. A high-fidelity FEA solve for every candidate would require more time, software interaction, and computation than a resource-constrained student project could support. A practical method was needed to use inexpensive models for broad exploration and reserve manual FEA for a small, traceable candidate set.

The second problem concerned evidence integrity. A useful workflow must preserve the connection between the engineering definition, design variables, calculated metrics, generated geometry, load and support regions, and validation record. A geometry generator must not report success when a required boundary region is missing. A stale response must not be attached to a new design. A CAD mass must not be presented as proof of structural performance. A solver report must not be accepted when its input load is incorrect.

These problems lead to a narrow research gap. The methods used by the project are established, but they are rarely presented together as a small, inspectable reference workflow that works on an ordinary engineering workstation, supports several marine component archetypes, contains failure, and separates analytical, CAD, and FEA evidence.

1.5 Aim

The project aimed to develop and evaluate an auditable, resource-conscious workflow for the parameterized mass optimization and validation preparation of marine structural components.

1.6 Specific Objectives

The specific objectives were to:

1. define part parameters, units, bounds, baselines, materials, loads, constraints, fixed geometry, and semantic boundary roles in inspectable source files;
2. develop first-order analytical models for mass, stress, yield-based factor of safety, and displacement;

3. explore each bounded design space with seeded Latin-hypercube sampling;
4. implement a scaled, bounded, multi-start Nelder–Mead search;
5. preserve invalid candidates and unavailable evaluators as controlled failed records instead of terminating an optimization campaign;
6. implement a versioned file protocol between the external Python application and an Autodesk Fusion add-in;
7. generate native Fusion geometry, STEP files, optional STL files, physical properties, and semantic boundary metadata;
8. test each native CAD generator over its baseline and 20 seeded design vectors;
9. prepare baseline and finalist packages for manual Fusion Static Stress studies;
10. demonstrate the complete intended method with the bracket proof of concept;
11. show how the same software architecture extends to a lifting padeye and stabilizer fin; and
12. state the engineering evidence required before any manufacturing or service decision.

1.7 Research Questions

The study was guided by six questions:

1. Can one part-aware software architecture represent and process a bracket, padeye, and stabilizer without bracket-only assumptions in shared workflow code?
2. Can seeded Latin-hypercube sampling and multi-start bounded Nelder–Mead search find lower-mass, analytically feasible candidates for all three parts?
3. Can the Fusion bridge rebuild valid native CAD and identify every required semantic boundary role across bounded geometry changes?
4. How closely do analytical mass estimates agree with the Fusion CAD mass for each baseline and selected finalist?
5. What evidence is still required before the analytical candidates can be treated as validated structural designs?
6. What lessons does the bracket proof of concept provide for future padeye and stabilizer FEA studies?

1.8 Case-Study Components

The three components represent different load paths and geometry classes. Table 1.1 gives their principal roles.

Table 1.1: Case-study components and the engineering behavior represented

Part	Material and load	Parameterized features	Principal behavior represented
Bracket	Aluminum 6061-T6; 1.4715 kN equip- ment envelope	Baseplate, ribs, and fil- let	Concentrated equip- ment load transferred through a ribbed plate to four mounting holes
Padeye	S275 steel; 14.715 kN pin load envelope	Doubler, lug, neck, gussets, and fillet	Pinned lifting load transferred through a lug and reinforcement into a supporting plate
Stabilizer	Aluminum 6061-T6; 11.808 kN pre- scribed resultant	Skin, spars, ribs, in- sert, and root fillet	Distributed transverse load carried by a fixed-envelope internal structure to a root flange

The bracket is the completed proof-of-concept case. It preserves the original project demonstration from a parameterized definition through analytical search, native CAD, and a manually prepared Static Stress study. The padeye and stabilizer use the same architecture and have analytical models, native CAD generators, semantic boundary roles, sweep evidence, and validation packages. Their final manual FEA campaigns remain future validation work.

1.9 Research Contribution

The contribution is an implementation-and-evaluation contribution, not a new optimization algorithm. It has four main parts.

1.9.1 Part-Aware Engineering Definitions

Each component is defined by one unit-aware manifest. The manifest contains exact parameter names, declared units, lower and upper bounds, baseline values, material properties, loads, constraints, fixed geometry, and semantic boundary roles. Evaluation, optimization, logging, validation packaging, and Fusion generation use the same

definition. This design reduces the risk that one layer applies a different load, bound, or unit.

1.9.2 Failure-Contained Optimization

The optimizer scales each variable to the interval from zero to one, clips trial points at the bounds, applies explicit penalties, caches repeated evaluations, and records every unique evaluation. Invalid candidates and unavailable evaluators return a controlled high objective. One bad candidate does not terminate the complete search. The final candidate selection filters explicitly for feasibility instead of assuming that the smallest penalized objective must pass every constraint.

1.9.3 Inspectable CAD Bridge

The bridge separates two Python environments. The external environment can use the full project dependency set but cannot directly run inside Fusion. The Fusion environment has access to the native design API but should not rely on external scientific packages. Versioned JSON request and response files connect the environments. Atomic replacement reduces partial-read risk, while the run identifier and iteration prevent a stale response from being accepted as a new result.

1.9.4 Semantic Boundary Evidence

Each generated CAD model identifies engineering regions such as `fixed_support`, `equipment_load`, `pin_bearing`, `pressure_band_1`, and `tip_monitor`. The metadata records geometric selectors and face signatures. This method does not guarantee permanent CAD face identity, but it makes the intended load and support meaning explicit and gives the engineer a checkable starting point for FEA.

1.10 Significance of the Study

The work has engineering, educational, and software significance. From an engineering perspective, the workflow can reduce the number of designs that require manual CAD and FEA attention. It also shows where a low-cost model is adequate for ranking and where higher-fidelity work is necessary.

From an educational perspective, the workflow makes hidden assumptions visible. A student can inspect the material, load derivation, bounds, equations, penalties, request files, response files, and validation checklists. The system therefore supports discussion of model credibility, not only the display of an optimized shape.

From a software perspective, the project shows how a small system can remain part-aware across the registry, analytical evaluation, optimizer, logger, dashboard, bridge,

and CAD generator. It also shows the value of returning a controlled partial result. The absence of a solver result remains visible instead of being hidden behind a successful CAD operation.

The zero-cash-cost execution is also important. The project used existing student hardware, educational software access, open-source Python packages, local file exchange, and university supervision. This arrangement makes the method accessible without a cloud-computing budget. It does not imply that the hardware, software access, electricity, internet, supervision, and student labor have no economic value. Chapter 5 distinguishes direct expenditure from these in-kind resources.

1.11 Scope

1.11.1 Implemented Software Scope

The implemented scope contains:

- three registered part definitions;
- first-order linear-elastic screening models;
- six continuous bracket variables and nine variables for each of the padeye and stabilizer;
- mass minimization with factor-of-safety and displacement penalties;
- random and Latin-hypercube sampling;
- bounded multi-start Nelder–Mead optimization;
- CSV and JSON run records;
- a Streamlit results dashboard;
- native Fusion generators for all three parts;
- STEP and optional STL export;
- semantic boundary metadata;
- baseline plus 20-vector CAD generator sweeps;
- baseline plus five-finalist validation packages; and
- a bracket Static Stress setup demonstration.

1.11.2 Engineering Scope

The analytical models estimate mass, a governing stress, yield-based factor of safety, and displacement. The bracket model uses an equivalent cantilever representation. The

padeye model compares net-section, bearing, shear-out, and lateral-bending stress estimates. The stabilizer model uses a root-section bending and shear representation under a prescribed distributed load.

These models support screening and ranking. They do not resolve local three-dimensional stress, fastener preload, weld detail, pin contact, geometric or material non-linearity, shell buckling, fatigue damage, corrosion, impact, vibration, fluid–structure interaction, or hydrodynamic pressure prediction.

1.11.3 Evidence Scope

The report uses four evidence levels:

1. **software evidence**, from automated offline tests and controlled command-line runs;
2. **analytical evidence**, from the declared first-order models;
3. **CAD evidence**, from native geometry, mass, volume, exports, and boundary-role sweeps; and
4. **FEA evidence**, which requires checked inputs, equilibrium, mesh convergence, and correctly recorded solver outputs.

Only the first three levels are complete for all parts. The bracket has a manual FEA setup report, but its saved load is incorrect by a factor of one thousand. Its numerical solver results are therefore excluded from structural acceptance.

1.12 Assumptions

The principal assumptions are:

1. the declared material properties represent homogeneous, isotropic, linear-elastic material during screening;
2. the bracket design envelope uses a 50 kg equipment mass and a dynamic amplification factor of 3;
3. the padeye design envelope uses a 500 kg working mass and a dynamic amplification factor of 3;
4. the stabilizer load uses a prescribed lift coefficient, planform area, speed, seawater density, dynamic factor, and spanwise load distribution;
5. the declared parameter bounds prevent most grossly invalid geometry but do not prove manufacturability;

6. the factor-of-safety constraint uses material yield strength and does not represent fatigue or buckling;
7. the displacement limits are project criteria, not class-approved service limits; and
8. the native CAD mass provides a geometry consistency check but does not validate stress or displacement.

1.13 Limitations and Delimitations

The study does not claim a global optimum. Nelder–Mead is a local method, the number of starts is finite, and a 60-point sample remains sparse in a nine-dimensional space. The defensible claim is the best feasible design found within the declared parameterized search.

The project does not complete a formal multi-fidelity optimization loop. Analytical predictions are not automatically corrected by FEA. The current workflow is tiered: analytical screening comes first, CAD verification comes second, and manual FEA comes last.

The project does not establish classification compliance, statutory compliance, fabrication readiness, or service safety. A later design stage must select the governing rules and load combinations, verify material and fabrication data, perform calibrated FEA, address fatigue, buckling, corrosion, contact, fasteners, and welds, and obtain qualified engineering review.

1.14 Safety and Claim Boundary

Every analytical value in this report is a screening estimate. A design with a reported factor of safety above 2.5 has passed the implemented yield-based screening equation. It has not passed all relevant marine failure modes. A successful CAD generation result proves only that the requested geometry and artifacts were produced under the generator acceptance checks.

A Fusion study is not credible only because it solved. The material, load, support, contacts, mesh, equilibrium, result location, and convergence must also be correct. This principle is demonstrated by the bracket report: Fusion completed a solve, but inspection found that the study used 1.472 N instead of 1472 N. The project rejects those numerical results.

1.15 Organization of the Report

Chapter 2 reviews constrained structural optimization, derivative-free search, sampling, tiered model fidelity, CAD–CAE integration, simulation intent, marine applications, and

FEA credibility. Chapter 3 describes the engineering definitions, analytical models, optimizer, data contracts, CAD bridge, verification method, and bracket FEA procedure. Chapter 4 presents the analytical, software, and CAD results and discusses the FEA quality-control finding. Chapter 5 gives the zero-cash-cost budget, work breakdown, 13-week workplan, Gantt chart, milestones, responsibilities, and risk plan. Chapter 6 answers the research questions and gives the conclusions and recommendations. The appendices provide detailed parameter tables, reproduction procedures, schema examples, and the validation checklist.

Chapter 2

Literature Review

2.1 Chapter Overview

This chapter reviews the established methods on which NEMO is based. The review covers constrained structural optimization, derivative-free search, Nelder–Mead, variable scaling, Latin-hypercube sampling, tiered model fidelity, CAD–CAE integration, simulation intent, marine structural applications, and computational-model credibility. It then relates these topics to the bracket, padeye, and stabilizer case studies and defines the limited research gap addressed by this project.

2.2 Structural Optimization as a Constrained Search

Structural optimization formalizes the selection of design variables to improve an objective while satisfying constraints. An objective can represent mass, cost, compliance, displacement, stress, fatigue damage, environmental impact, or a weighted combination of measures. Constraints can represent strength, stiffness, stability, vibration, geometry, fabrication, inspection, or regulatory limits. The problem is written in the general form

$$\underset{\mathbf{x}}{\text{minimize}} \quad f(\mathbf{x}), \tag{2.1}$$

$$\text{subject to} \quad g_j(\mathbf{x}) \leq 0, \quad j = 1, \dots, m, \tag{2.2}$$

$$h_k(\mathbf{x}) = 0, \quad k = 1, \dots, p, \tag{2.3}$$

$$\mathbf{l} \leq \mathbf{x} \leq \mathbf{u}, \tag{2.4}$$

where $\mathbf{x}$ is the design vector, f is the objective, g_j and h_k are inequality and equality constraints, and $\mathbf{l}$ and $\mathbf{u}$ are variable bounds.

Structural optimization is commonly divided into size, shape, and topology optimization. Size optimization changes dimensions such as thickness and section area. Shape optimization changes geometric boundaries while retaining a general feature ar-

range. Topology optimization changes the distribution or connectivity of material inside a design region. These classes can overlap, but the distinction helps define what an optimization claim means.

NEMO performs size and bounded parametric shape optimization. The number and general arrangement of major features remain fixed. The optimizer changes dimensions such as plate thickness, rib height, lug width, spar position, and fillet radius. It does not remove arbitrary material or create a new topology. This restriction keeps every candidate compatible with an intended load path, a native CAD generator, and predefined load and support roles.

The project uses mass as the principal objective. This choice is easy to measure and compare, but it does not make mass the only design criterion. A low-mass candidate is useful only if its constraints pass and its omitted failure modes are later checked. In marine practice, this qualification is especially important because cyclic loading, corrosion, fabrication detail, and uncertain operational loads can govern a design that passes one static yield check.

2.3 Derivative-Free Optimization

Many engineering analyses appear as black-box functions to the optimizer. The optimizer supplies dimensions and receives calculated outputs, but it does not have exact derivatives of those outputs. CAD regeneration, meshing changes, solver tolerances, discrete feature behavior, and failed candidates can also make numerical gradients unreliable.

Direct-search and derivative-free methods use objective values without requiring analytical gradients. Kolda, Lewis, and Torczon reviewed the development of direct-search methods and distinguished methods with formal convergence properties from more heuristic searches [3]. Their review shows that derivative-free operation is not the same as assumption-free operation. The designer must still scale variables, handle bounds, define constraints, select termination criteria, and evaluate sensitivity to the starting point.

Derivative-free methods can be attractive when each function evaluation is moderately expensive and the number of variables is limited. They are less attractive when the dimension is very high or when gradients can be calculated cheaply and reliably. The three NEMO cases contain six or nine continuous variables. Their analytical evaluations are inexpensive, while the wider workflow includes potential discontinuities from geometry checks. This combination supports the use of a simple derivative-free method for the reference implementation.

2.4 The Nelder–Mead Simplex Method

Nelder and Mead introduced the simplex method in 1965 [4]. In an n -dimensional space, the method maintains $n + 1$ vertices. It evaluates and ranks these vertices, calculates the centroid of the better vertices, and attempts to replace the worst vertex. The standard trial operations are reflection, expansion, contraction, and shrink.

If $\mathbf{x}_h$ is the worst vertex and $\bar{\mathbf{x}}$ is the centroid of the remaining vertices, a reflected point is

$$\mathbf{x}_r = \bar{\mathbf{x}} + \alpha(\bar{\mathbf{x}} - \mathbf{x}_h), \quad (2.5)$$

where α is the reflection coefficient. A sufficiently good reflection can lead to expansion:

$$\mathbf{x}_e = \bar{\mathbf{x}} + \gamma(\mathbf{x}_r - \bar{\mathbf{x}}). \quad (2.6)$$

An unsuccessful reflection can cause contraction toward the centroid. When contraction also fails, the simplex shrinks toward its best vertex. The standard coefficients used by NEMO are $\alpha = 1$, $\gamma = 2$, $\rho = 0.5$, and $\sigma = 0.5$.

The method has several practical advantages. It is compact, does not require a gradient, and is easy to explain and implement. It also has important limits. Lagarias *et al.* established convergence results under restricted conditions and discussed cases where the method can approach a non-minimizer [5]. Nelder–Mead is a local search. Its final design can depend on the initial simplex, variable scaling, bounds, penalties, and the shape of the objective landscape.

These limitations affect the language of this report. One local run cannot support a claim of global optimality. NEMO uses a seeded design-space sample and three starts to reduce dependence on the configured baseline. The report uses the phrase “best feasible design found within this parameterized search.” This wording is more defensible than “global optimum.”

Table 2.1: Nelder–Mead operations and their purpose in the implemented search

Operation	Coefficient	Purpose
Reflection	$\alpha = 1$	Tests a point across the centroid from the current worst vertex.
Expansion	$\gamma = 2$	Extends a successful reflected step when it improves the best objective.
Contraction	$\rho = 0.5$	Tests a shorter step when reflection does not give adequate improvement.
Shrink	$\sigma = 0.5$	Moves all non-best vertices toward the best point after failed trial steps.

2.5 Variable Scaling, Bounds, and Constraint Penalties

Engineering variables can differ greatly in magnitude and unit. A plate thickness may be 4 mm, a chordwise spar position may be 30%, and a root insert length may be 180 mm. If an optimizer works directly in these physical coordinates, its simplex geometry and trial step sizes can be dominated by the numerically largest variables.

NEMO maps each physical variable x_i to the scaled coordinate

$$z_i = \frac{x_i - l_i}{u_i - l_i}, \quad 0 \leq z_i \leq 1, \quad (2.7)$$

where l_i and u_i are the lower and upper bounds. The evaluator performs the inverse mapping before it uses the engineering equations. Trial points are clipped at the scaled bounds.

Scaling gives each coordinate the same numerical range. It does not make the physics equally sensitive. A small thickness change can alter both mass and section properties strongly, while a similar scaled change in a fillet radius may affect only one stress factor in the analytical model. Sensitivity remains a property of the model.

Nelder–Mead does not directly enforce general nonlinear constraints in this implementation. NEMO therefore uses a penalty objective. If $m(\mathbf{x})$ is mass, $F(\mathbf{x})$ is factor of safety, $F_{\min}$ is the minimum factor of safety, $\delta(\mathbf{x})$ is displacement, and $\delta_{\max}$ is the displacement limit, the objective is

$$J(\mathbf{x}) = m(\mathbf{x}) + w \left[\left(\frac{\max(0, F_{\min} - F(\mathbf{x}))}{F_{\min}} \right)^2 + \left(\frac{\max(0, \delta(\mathbf{x}) - \delta_{\max})}{\delta_{\max}} \right)^2 \right], \quad (2.8)$$

with $w = 100$ for the current parts.

A feasible design has $J = m$. An infeasible design has a higher objective according to its normalized violations. Penalties simplify the search, but their weight affects ranking. A weak penalty can make an infeasible light design more attractive than a feasible heavier design. A large penalty can create a steep landscape near the boundary. The final selection must therefore filter explicitly for feasibility. The stabilizer case in Chapter 4 demonstrates why this check is necessary.

2.6 Latin-Hypercube Sampling

McKay, Beckman, and Conover introduced Latin-hypercube sampling as a stratified method for computer experiments [6]. For each variable, an N -point Latin hypercube samples once from each of N equal-probability intervals. Independent permutations combine the one-dimensional samples into N multi-dimensional points.

Compared with an unstructured random sample of the same size, a Latin hypercube improves one-dimensional stratification. It does not guarantee dense coverage of every interaction, especially in high dimensions. Sixty points remain sparse in a nine-dimensional space. The method is therefore best viewed here as an economical exploration step.

NEMO uses 60 samples per part with seed 42. The fixed seed makes the campaign repeatable. Sampling serves three purposes:

1. it checks whether calculated outputs change in a physically sensible direction over the bounds;
2. it identifies low-mass feasible candidates that can start local search; and
3. it reduces reliance on one baseline-start Nelder–Mead run.

Sampling also provides diagnostic evidence. A high failure rate can reveal poor parameter bounds or a fragile geometry generator. An unexpectedly high feasibility rate can reveal an over-conservative baseline or a screening model that does not activate the declared constraints.

2.7 Tiered Model Fidelity

Engineering optimization often combines models with different cost and accuracy. Perherstorfer, Willcox, and Gunzburger review multi-fidelity methods that combine inexpensive low-fidelity models with expensive high-fidelity models [7]. Formal multi-fidelity optimization uses the high-fidelity model inside the optimization strategy to improve prediction, selection, or convergence.

NEMO does not yet implement such a closed loop. The analytical model guides the search. Native CAD verifies geometry and mass for selected designs. Manual FEA follows as a separate validation stage. The correct description is a *tiered-fidelity workflow*. The analytical optimum is not automatically corrected by FEA.

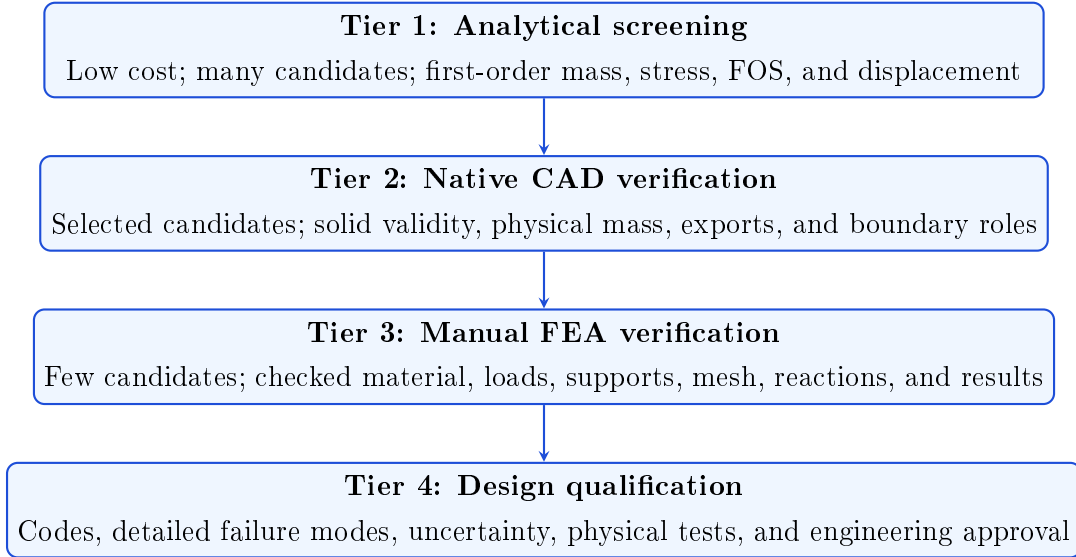


Figure 2.1: Tiered evidence model used by the project

The distinction between the tiers protects the report from a common error: treating agreement on one quantity as complete validation. Close agreement between analytical and CAD mass shows that the volume approximations represent the generated geometry reasonably well. It does not show that the analytical stress model represents local three-dimensional stress.

2.8 CAD–CAE Integration

Automated structural optimization requires dependable communication between geometry, analysis, and optimization. Park and Dang coupled commercial CAD and CAE software through scripting, programming interfaces, and metamodels [1]. Their work shows how integration can reduce repetitive work and analysis cost. It also introduces an additional source of error when a surrogate model approximates the high-fidelity response.

Li *et al.* developed a unified parametric representation based on extended voxels. Their method supports a closed loop among design, simulation, and optimization while carrying design and simulation meaning [8]. This work confirms that automated CAD–CAE optimization is an established research area. NEMO does not claim novelty from automation alone.

The project instead uses a modest file-based architecture. External Python handles engineering definitions, optimization, run records, and validation packages. Fusion handles native CAD and physical properties. Versioned JSON files connect the two en-

vironments. This approach is slower than an in-process call but has useful properties:

- a request and response can be inspected with an ordinary text editor;
- the Fusion environment does not need third-party optimization packages;
- a failed request can preserve a controlled error record;
- the protocol can retain compatibility with the original bracket proof of concept; and
- the same external application can later support another CAD or FEA backend.

The architecture also has limits. A single shared request directory serializes CAD work. Polling adds latency. Local file locking requires retry behavior. The bridge cannot make an unavailable solver API available. These are accepted trade-offs for an inspectable student-scale implementation.

2.9 Simulation Intent and Semantic Boundary Roles

Geometry changes can invalidate an analysis even when the regenerated solid looks correct. A load attached to a low-level face identifier can disappear or move after a feature changes the boundary representation. A reliable parametric workflow must preserve why a region exists and how it is used.

Nolan *et al.* define simulation intent as the high-level modeling and idealization decisions required for fit-for-purpose analysis [2]. Their work associates analysis meaning with regions and interfaces rather than relying only on unstable topology.

NEMO applies the same principle in a smaller implementation. The bracket declares `fixed_support` and `equipment_load`. The padeye declares `fixed_support` and `pin_bearing`. The stabilizer declares `fixed_support`, four pressure bands, and `tip_monitor`. The Fusion generator selects faces with part-specific geometric rules and records area, centroid, normal, bounding box, and cylinder data where available.

Semantic roles improve traceability, but they do not remove the need for human inspection. A broad geometric selector can match an unintended face. A topology change can divide one intended region into several faces. The manual FEA procedure must display and confirm every selected face before solving.

2.10 Marine Structural Optimization

Marine structural optimization must represent realistic loads, supports, and failure modes. Gentils, Wang, and Kolios optimized an offshore wind support structure with parametric FEA and a genetic algorithm [9]. Their constraints included stress, displacement, vibration, buckling, and fatigue. The study reported a 19.8% mass reduction for

the complete support structure.

Zheng, Li, and Xiao later used sensitivity analysis, surrogate models, and a genetic algorithm for jacket structures [10]. Their method addressed natural frequency, displacement, stress, buckling, and later fatigue checks. Wang, Mantey, and Zhang developed a fast parametric FEA and genetic-algorithm tool for jacket structures [11]. These studies show that marine structural optimization commonly requires several load cases and constraints beyond static yield.

The literature provides two lessons for NEMO. First, mass reduction must remain subordinate to the governing engineering constraints. Second, the selected design can change when supports and loads change. The project therefore treats its load cases as declared engineering envelopes, not as complete vessel or offshore design cases.

Table 2.2: Relationship between selected literature and the NEMO implementation

Source area	Established lesson	Use in NEMO
Direct search [3–5]	Derivative-free local search is useful but needs scaling, stopping rules, and careful claims.	Scaled bounded variables, three starts, and “best design found” wording.
Computer experiments [6]	Stratified samples improve economical exploration.	Sixty seeded Latin-hypercube samples per part.
Multi-fidelity methods [7]	Models of different cost require a defined relationship.	Tiered analytical, CAD, manual FEA, and qualification evidence.
CAD–CAE integration [1,8]	Automation must preserve data and analysis meaning.	Versioned JSON bridge, native CAD, and explicit partial responses.
Simulation intent [2]	Boundary meaning must survive geometry change.	Semantic roles and geometric face signatures.
Marine optimization [9–11]	Mass is one objective among strength, stiffness, stability, dynamics, and fatigue.	Static screening followed by a stated list of required detailed checks.

2.11 Bracket Structural Behavior

Autodesk lists brackets among components suited to linear Static Stress analysis when the assumptions of small deformation, unchanged load direction, and linear elastic material behavior hold [12]. The apparent simplicity of a bracket can be misleading. Local stress depends on load introduction, fastener restraint, contact, rib geometry, fillet radius, plate flexibility, and weld detail.

The NEMO bracket is an equipment-foundation archetype with a baseplate, four

mounting holes, two ribs, and rib-root fillets. Its analytical model uses an equivalent cantilever relation. This model captures the broad effects of section depth, plate thickness, rib thickness, and load eccentricity. It does not resolve bolt bearing, fastener preload, prying, weld-toe stress, local contact, or three-dimensional restraint.

The bracket is useful as a proof of concept because the geometry is easy to recognize and the load path is easy to explain. It is not useful because it is free from modeling risk. The proof-of-concept FEA report later showed that a three-order-of-magnitude input error can survive through a completed solver run. This finding supports a workflow that checks input values and reaction balance before it discusses result contours.

2.12 Padeyes and Pinned Connections

Padeyes transfer lifting force through a pin hole into a lug, reinforcement, welds, and supporting structure. Potential governing effects include pin-bearing stress, net-section tension, shear-out, bending, local contact, lug-root stress, gusset load transfer, and weld stress. Clearance, load angle, out-of-plane load, and local geometry can strongly affect the stress field.

Soh and Soh compared simplified two-dimensional padeye models with a three-dimensional finite-element solution [13]. Their work showed that model selection affects stress prediction and can lead to conservative overdesign. Liu, Zhou, and Tan used FEA to study padeye stress and deformation and then formulated an optimization problem to reduce steel use [14]. These studies support the use of a simplified model for early screening, provided later FEA represents the important three-dimensional behavior.

Pedersen examined the stress concentration around a pin-loaded hole and showed the influence of contact modeling and local geometric refinement [15]. This result is directly relevant to the NEMO padeye. Its current analytical model compares several nominal stress modes, but it does not model a separate pin, clearance, nonlinear contact, or a weld group. A later FEA campaign must decide whether a distributed bearing load is adequate for screening or whether explicit pin contact is required.

Marine lifting appliances also require a governing rule set. DNV-ST-N001 addresses marine operations and associated structural considerations [16]. Lloyd's Register publishes a Code for Lifting Appliances in a Marine Environment that covers design, materials, fabrication, testing, marking, survey, and documentation [17]. NEMO does not claim compliance with either source. They are cited to show the wider qualification work that lies beyond a single yield-based optimization.

2.13 Stabilizer Fins and Prescribed Hydrodynamic Loads

The stabilizer differs from the bracket and padeye because its load is distributed over an external lifting surface. The project retains a NACA 0015 section. Abbott, von Doenhoff, and Stivers documented NACA airfoil families and aerodynamic data in NACA Report 824 [18]. The use of a known profile makes the geometric definition reproducible, but it does not supply a complete marine pressure field.

The NEMO stabilizer keeps its span, root chord, tip chord, sweep, trailing edge, root flange, and rib stations fixed. It changes skin thickness, two spar positions, two spar thicknesses, rib thickness, root-insert dimensions, and root fillet radius. Its prescribed resultant is calculated from a simplified lift expression and distributed through four spanwise weights.

This setup supports internal structural sizing under one declared load envelope. It does not optimize hydrodynamic efficiency. It omits vessel motions, unsteady flow, free-surface effects, ventilation, cavitation, actuator behavior, pressure reversal, impact, and fluid–structure interaction. A later design stage must derive the pressure from an accepted hydrodynamic method, CFD, model testing, or full-scale data.

2.14 Linear Static Finite Element Analysis

Autodesk describes a Static Stress study as appropriate when deformation is small, load direction does not change significantly, material properties are constant, and the response remains linear elastic [12]. A complete setup requires geometry, materials, constraints, loads, contacts, mesh, solution, and result review [19].

These requirements are coupled. A correct material with a wrong load gives a wrong result. A correct load on the wrong face can change the load path. An overly rigid support can reduce deformation and create a local stress singularity. A coarse mesh can underpredict a local stress. A maximum contour value can occur at a mathematical singularity that is unsuitable for direct acceptance.

Reaction forces provide a basic equilibrium check. Autodesk notes that support reactions should oppose the total applied load [20]. For the bracket, a correctly configured 1472 N downward load should produce a vertical support reaction close to 1472 N, subject to sign convention and small numerical differences. A reaction near 1.472 N would immediately expose the factor-of-one-thousand load error.

Mesh convergence is also necessary. Autodesk explains that coarse meshes commonly underpredict stress and that stress is more mesh-sensitive than displacement [21]. Comparison should use the same geometric location across refinements. A credible record should report the global mesh, local refinements, element order, node and element count,

the monitored quantity, and the change between successive meshes.

2.15 Verification, Validation, and Model Credibility

Verification asks whether equations and software were solved or implemented correctly. Validation asks whether the model represents the real system well enough for its intended use. These questions are related but distinct. A correctly implemented simplified model can still be unsuitable for a service decision.

ASME V&V 10 provides a framework and terminology for verification, validation, and uncertainty quantification in computational solid mechanics [22]. NASA-STD-7009B emphasizes intended use, acceptance criteria, input pedigree, uncertainty, sensitivity, verification, validation, and communicated credibility [23]. The principles are relevant to NEMO even though the project does not fall under NASA governance. Model credibility depends on the decision being supported.

Table 2.3: Evidence levels and the questions they answer

Evidence level	Primary question	Representative checks
Software verification	Did the code implement its declared contracts and calculations consistently?	Unit tests, bounds, baseline feasibility, failure handling, logging schema, repeatable seeds.
CAD verification	Did the requested dimensions create a usable native solid and required artifacts?	One connected solid, positive volume, STEP export, boundary roles, physical mass.
FEA verification	Was the numerical study configured and solved correctly?	Material, load, support, contact, reaction balance, mesh convergence, result location.
Physical validation	Does the model represent measured component behavior adequately for its intended use?	Proof load, strain, displacement, failure mode, repeatability, uncertainty.
Design qualification	Does the complete design satisfy governing service requirements?	Design codes, load combinations, fatigue, buckling, corrosion, fabrication, inspection, competent approval.

The table shows why a project can complete software and CAD verification without completing structural validation. It also explains why a single FEA contour is not final proof. The bracket report is retained because it documents the setup and exposes a process weakness. Its incorrect numerical result is not promoted to a higher evidence level.

2.16 Literature Synthesis and Research Gap

The literature establishes every main method used by NEMO. Nelder–Mead provides a compact derivative-free local search [4, 5]. Direct-search literature explains the need for careful implementation and claims [3]. Latin-hypercube sampling provides a repeatable exploration stage [6]. Multi-fidelity literature explains the value and limits of combining models of different cost [7]. CAD–CAE research identifies integration and simulation intent as central problems [1, 2, 8]. Marine optimization studies show that useful lightweighting depends on several constraints and credible load models [9–11]. Computational-mechanics guidance requires a declared intended use and an evidence chain [22, 23].

No single method in NEMO is new. The reviewed literature also contains more advanced automated loops, global optimizers, surrogate models, and high-fidelity marine applications. The project therefore does not claim to be the first marine optimizer or the first CAD–CAE automation system.

The limited gap addressed here is practical integration at student-project scale. The reviewed sources do not directly provide a transparent, part-manifest-driven workflow that combines unit-aware definitions, first-order screening, seeded sampling, scaled bounded multi-start search, failure-contained evaluation, versioned atomic file exchange, native Fusion CAD, semantic boundary metadata, validation packages, and explicit separation of analytical, CAD, and FEA evidence across a bracket, padeye, and stabilizer.

NEMO addresses that implementation gap as an auditable reference workflow. Its value lies in exposing each assumption and handoff, reducing repetitive candidate work, and preserving enough evidence for a supervisor or engineer to check the result. The next chapter describes how the workflow was implemented and evaluated.

Chapter 3

Methodology and System Development

3.1 Chapter Overview

This chapter describes the research design, software architecture, data contracts, engineering definitions, analytical equations, design-space exploration, optimization method, native CAD generation, verification campaign, and manual FEA procedure. The bracket is used as the complete proof of concept. The final sections define how the same method will be applied to the padeye and stabilizer.

3.2 Research Design

The project used a design-and-evaluate research method. It created an engineering software artifact, tested the artifact, applied it to three case studies, and evaluated the resulting analytical and CAD evidence against declared requirements. The work did not use experimental human subjects or a statistical population. Its evidence came from software behavior, analytical calculations, native CAD artifacts, and a manually configured solver report.

The work proceeded through four technical phases:

1. **Engineering definition:** define parameters, bounds, units, baselines, materials, loads, constraints, fixed geometry, and boundary roles.
2. **Search:** evaluate baselines, sample the design spaces, and run three bounded Nelder–Mead starts for each part.
3. **CAD verification:** rebuild selected designs in Fusion, calculate physical properties, export artifacts, and check boundary roles.
4. **Validation preparation:** package baselines and finalists, prepare manual Static Stress studies, and record the evidence required for engineering acceptance.

The phases follow the evidence tiers in Figure 2.1. A later tier can reject an earlier

candidate. For example, CAD failure can reject an analytically feasible design. Corrected FEA can reject a design that passes both analytical and CAD checks.

3.3 Functional and Non-Functional Requirements

Table 3.1 summarizes the main implementation requirements.

Table 3.1: Principal NEMO system requirements

ID	Requirement
FR1	Register the bracket, padeye, and stabilizer through one part-aware interface.
FR2	Read exact variables, units, bounds, baselines, materials, loads, constraints, fixed geometry, and boundary roles from part definitions.
FR3	Return mass, stress, factor of safety, displacement, volume, objective, status, and error fields through a consistent evaluation record.
FR4	Sample bounded variables with repeatable random seeds.
FR5	Run bounded multi-start Nelder–Mead search without requiring SciPy inside Fusion.
FR6	Preserve invalid evaluations with controlled failure status and a high objective.
FR7	Record one part per CSV run with stable column order.
FR8	Exchange versioned, human-readable requests and responses with NEMOBridge.
FR9	Correlate each response with both the run identifier and iteration.
FR10	Generate native Fusion CAD, physical mass, STEP, optional STL, and boundary metadata.
FR11	Create a validation package containing the baseline, selected finalists, Fusion requests, and a checklist.
FR12	Display recorded run results through a Streamlit dashboard.
NFR1	Use repository-relative configuration and no machine-specific credentials.
NFR2	Keep analytical, CAD, and FEA evidence visibly separate.
NFR3	Preserve schema-version-1 compatibility for the original bracket proof of concept.
NFR4	Avoid paid cloud services and operate through existing student and university resources.
NFR5	Do not let one invalid candidate terminate a complete optimization campaign.
NFR6	Do not fabricate unavailable FEA outputs.

3.4 System Architecture

The system contains four main layers. The Python package provides the CLI, part registry, samplers, analytical evaluators, optimizer, logger, bridge client, and validation-package generator. The JSON part definitions provide the engineering source of truth. NEMOBridge runs inside Fusion and provides native CAD generation. The dashboard reads completed run records.

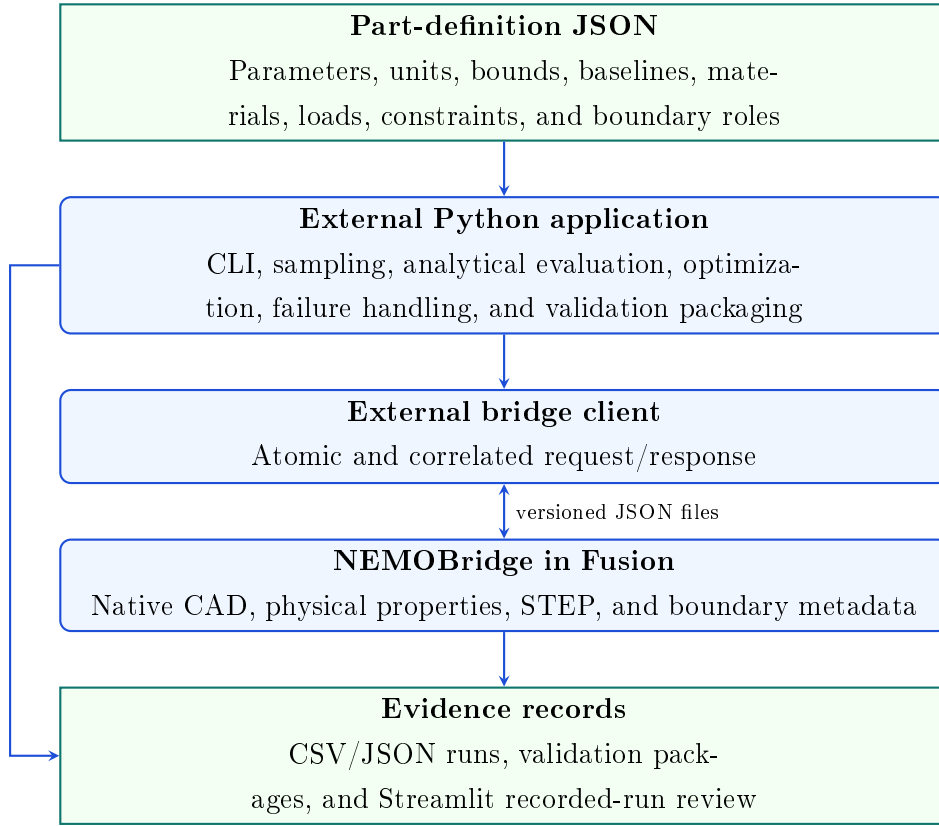


Figure 3.1: NEMO system architecture

The external Python package is installed from `src/nemo`. It has access to the project dependencies and file system. Fusion uses an embedded Python environment with its own application lifecycle. Autodesk defines scripts and add-ins through a manifest and `run` and `stop` entry points [24]. The file bridge avoids importing the external package inside Fusion.

3.5 Repository and Evidence Organization

The repository separates source, generated run data, validation packages, documentation, reports, and the preserved proof of concept. The important locations are:

- `src/nemo`: installable Python package and CLI;
- `src/nemo/parts/definitions`: authoritative part manifests;

- `src/nemo/parts/analytical.py`: screening equations;
- `fusion_addin/NEMOBridge`: Fusion generator and bridge code;
- `data/runs`: generated optimization logs and the live channel;
- `reports`: validation packages and report sources;
- `docs`: engineering assumptions and interface contracts; and
- `demos/bracket-proof-of-concept`: original demonstration, instructions, and historical validation packages.

The separation prevents generated artifacts from being treated as source code. It also keeps the bracket demonstration available without allowing legacy schema assumptions to dominate the current multi-part workflow.

3.6 Part-Definition Contract

Each part manifest declares the fields needed by both the external package and the Fusion generator. Parameter names are exact contract keys. Units are explicit. Most variables use millimeters, while the stabilizer spar positions use percent chord. The evaluator converts millimeters to meters before using SI equations and converts outputs to MPa and millimeters for reporting.

The manifest includes:

- part identifier and display name;
- parameter name, label, unit, lower bound, baseline, and upper bound;
- material density, yield strength, Young’s modulus, and Poisson ratio;
- design-load derivation and applied value;
- factor-of-safety and displacement limits;
- fixed dimensions and feature counts;
- supported evaluation modes; and
- semantic boundary roles required from the CAD generator.

This contract makes a part change a multi-layer change. A new variable must be available to the analytical model, optimizer, logger, validation package, CAD generator, and any boundary selector affected by its geometry.

3.7 Bracket Engineering Definition

The bracket represents a foundation for a 50 kg auxiliary unit. The design load uses a project dynamic factor of 3:

$$F_d = mg\gamma_d = 50(9.81)(3) = 1471.5 \text{ N.} \quad (3.1)$$

The dynamic factor is an engineering envelope. It is not a modal, shock, or vessel-motion result. The analytical calculation retains 1471.5 N. The current Fusion input dialog does not accept the decimal force value used in this workflow. The manual study therefore uses the conservative integer value 1472 N, a difference of approximately 0.034%.

The project-defined Aluminum 6061-T6 properties are density 2700 kg/m³, yield strength 276 MPa, Young's modulus 68.9 GPa, and Poisson ratio 0.33. Table 3.2 gives the variables.

Table 3.2: Bracket design variables

Variable	Unit	Lower	Baseline	Upper
Baseplate length	mm	100	150	200
Baseplate width	mm	80	100	150
Baseplate thickness	mm	4	8	15
Rib height	mm	20	40	60
Rib thickness	mm	3	6	10
Fillet radius	mm	2	5	10

The fixed geometry includes four 10 mm mounting holes and two ribs. The screening acceptance criteria are $FOS \geq 2.5$ and $\delta \leq 0.5$ mm. These limits are project criteria and do not replace a governing marine rule set.

3.8 Bracket Analytical Model

All geometric dimensions enter the equations in meters. The approximate volume contains the rectangular baseplate, four-hole subtraction, two triangular ribs, and a fillet contribution:

$$V_b = LWt_b - n_h\pi \left(\frac{d_h}{2}\right)^2 t_b + n_r \frac{Lh_r t_r}{2} + n_r \frac{\pi r_f^2}{4} \max(0.35W, t_r). \quad (3.2)$$

The bracket mass is

$$m_b = \rho V_b. \quad (3.3)$$

The equivalent cantilever moment is

$$M_b = F_d e, \quad e = \max(0.03, 0.35L + 0.02). \quad (3.4)$$

The effective section modulus combines the baseplate and rib terms:

$$Z_{\text{eff}} = \frac{W t_b^2}{6} + 0.65 n_r \frac{t_r h_r^2}{6}. \quad (3.5)$$

The stress estimate includes a fillet-dependent concentration factor:

$$K_t = \max \left(1.08, 1.35 - \frac{r_f [\text{mm}]}{40} \right), \quad (3.6)$$

$$\sigma_b = \frac{M_b}{Z_{\text{eff}}} K_t. \quad (3.7)$$

The effective second moment of area is

$$I_{\text{eff}} = \frac{W t_b^3}{12} + 0.45 n_r \frac{t_r h_r^3}{36}, \quad (3.8)$$

and the tip-equivalent displacement is

$$\delta_b = \frac{F_d e^3}{3 E I_{\text{eff}}}. \quad (3.9)$$

The yield-based factor of safety is

$$FOS_b = \frac{\sigma_y}{\sigma_b}. \quad (3.10)$$

The equations preserve the expected broad trends. Increased plate or rib thickness increases mass and stiffness. Increased rib height strongly increases bending stiffness. Increased fillet radius reduces the assumed stress concentration. The model does not calculate bolt bearing, bolt preload, weld-group stress, weld-toe fatigue, local plate bending, contact, or three-dimensional load sharing.

3.9 Padeye Engineering Definition

The padeye represents a lifting point for a 500 kg working mass with a dynamic factor of 3:

$$F_d = 500(9.81)(3) = 14\,715 \text{ N}. \quad (3.11)$$

The project-defined S275 properties are density 7850 kg/m³, yield strength 275 MPa, Young's modulus 210 GPa, and Poisson ratio 0.30. The fixed pin-hole diameter is 50 mm, and two gussets reinforce the lug. The criteria are $FOS \geq 2.5$ and $\delta \leq 0.5 \text{ mm}$.

Table 3.3: Padeye design variables

Variable	Unit	Lower	Baseline	Upper
Base length	mm	220	260	320
Base width	mm	150	180	230
Base thickness	mm	8	14	22
Lug height	mm	140	180	230
Lug thickness	mm	12	20	30
Neck width	mm	80	100	140
Gusset height	mm	70	110	120
Gusset thickness	mm	6	10	18
Fillet radius	mm	8	15	25

3.10 Padeye Analytical Model

The volume model combines the doubler plate, a tapered lug and circular crown, the 50 mm hole subtraction, two triangular gussets, and a fillet approximation. This representation follows the native generator closely enough for a mass consistency check, but it is not an exact boundary-representation calculation.

The structural model calculates four candidate stresses. The net-section tension stress uses the remaining ligament at the hole and a fillet-dependent stress factor. The nominal bearing stress is

$$\sigma_{\text{bearing}} = \frac{F_d}{d_p t_l}, \quad (3.12)$$

where d_p is pin diameter and t_l is lug thickness. The shear-out estimate uses the two load paths between the hole and the outer edge:

$$\tau_{\text{out}} = \frac{F_d}{2A_{\text{shear}}}. \quad (3.13)$$

A lateral-bending estimate represents an assumed 25% side-load component. The reported padeye stress is

$$\sigma_p = \max \left(\sigma_{\text{net}}, \sigma_{\text{bearing}}, \sqrt{3} \tau_{\text{out}}, \sigma_{\text{lateral}} \right). \quad (3.14)$$

The displacement combines axial and lateral terms, and the factor of safety is $FOS_p = \sigma_y / \sigma_p$. This model ranks candidates by several relevant nominal modes. It does not model a separate pin, pin clearance, nonlinear contact, local peak stress, weld geometry, or fatigue. These limitations define the later FEA and design-code work.

3.11 Stabilizer Engineering Definition

The stabilizer retains a NACA 0015 outer envelope with 800 mm span, 500 mm root chord, 250 mm tip chord, 12-degree sweep, and 8 mm trailing edge. It has a 560 mm by 160 mm by 18 mm root flange and three ribs at 25%, 50%, and 75% span.

The project-defined Aluminum 6061-T6 properties match the bracket. The prescribed resultant follows

$$F_d = \frac{1}{2} \rho_w V^2 S C_L \gamma_d, \quad (3.15)$$

where seawater density is 1025 kg/m³, speed is 8 m/s, planform area is 0.3 m², lift coefficient is 0.8, and the dynamic factor is 1.5. Thus

$$F_d = \frac{1}{2} (1025) (8^2) (0.3) (0.8) (1.5) = 11\,808 \text{ N}. \quad (3.16)$$

Four pressure bands use relative weights 1.00, 0.93, 0.75, and 0.40. These are declared engineering assumptions, not CFD data. The screening criteria are $FOS \geq 2.5$ and tip displacement no more than 5 mm.

Table 3.4: Stabilizer design variables

Variable	Unit	Lower	Baseline	Upper
Skin thickness	mm	4	7	12
Front spar position	% chord	25	30	38
Rear spar position	% chord	58	65	72
Front spar thickness	mm	4	8	14
Rear spar thickness	mm	4	6	12
Rib thickness	mm	3	5	9
Root insert length	mm	120	180	260
Root insert thickness	mm	8	12	20
Root fillet radius	mm	12	25	40

3.12 Stabilizer Analytical Model

The volume model combines the two-sided skin, front and rear spar webs, three internal ribs, root insert, fixed flange, and root-fillet approximation. The NACA thickness equation estimates the local spar height at each chord position. The root section combines skin, spar, and insert contributions to the second moment of area.

The root bending moment uses a resultant centroid at 40% span:

$$M_r = 0.40 F_d b. \quad (3.17)$$

The model calculates a nominal root bending stress σ_b and shear stress τ . A root-fillet factor K_t modifies the bending term. The equivalent stress is

$$\sigma_s = \sqrt{(K_t \sigma_b)^2 + 3\tau^2}. \quad (3.18)$$

The tip displacement estimate is

$$\delta_s = \frac{F_d b^3}{8EI_{\text{eff}}}. \quad (3.19)$$

The model does not derive the pressure field. It also omits torsional deflection, shell buckling, panel crippling, local skin-spar stress, fatigue, fluid-structure interaction, and root-bearing detail. It therefore supports structural screening under a prescribed load, not hydrodynamic optimization.

3.13 Common Evaluation and Failure Handling

Each evaluator returns a structured record with volume, mass, maximum stress, factor of safety, maximum displacement, objective, status, and error. It first checks that the part exists, all required parameters are present, values are finite, values lie within bounds, and the requested evaluation mode is available.

Invalid inputs and unavailable evaluators return a failed record with objective 10^9 . The analytical optimizer can continue after such a record. A CAD-only response has partial status because it can contain valid mass and artifacts while structural metrics remain unavailable. The reserved `open_fea` mode deliberately returns a controlled failure. No Gmsh or CalculiX result is fabricated.

3.14 Design-Space Exploration

Each part used 60 Latin-hypercube samples with seed 42. The sampler operated in the scaled unit cube. Each variable was divided into 60 intervals. A random point was selected within each interval, and independently permuted variable columns formed the multi-dimensional vectors. The inverse scaling in Equation 2.7 produced physical values.

The sampler evaluated all points, retained failed records, and counted feasible candidates. A feasible candidate met both the factor-of-safety and displacement constraints. Promising feasible sampled designs supplied two of the three optimization starts.

3.15 Bounded Multi-Start Optimization

The pure-Python optimizer operated on scaled coordinates. Each part used three starts:

1. the configured baseline;

2. a promising low-mass feasible sample; and
3. a second promising feasible sample that provided another basin or parameter combination.

Each run permitted at most 80 simplex iterations. The optimizer converted each trial point to physical units, evaluated the part, cached repeated vectors, and appended each unique evaluation to the run log. The simplex stopped at the iteration limit or when the objective spread met the configured tolerance.

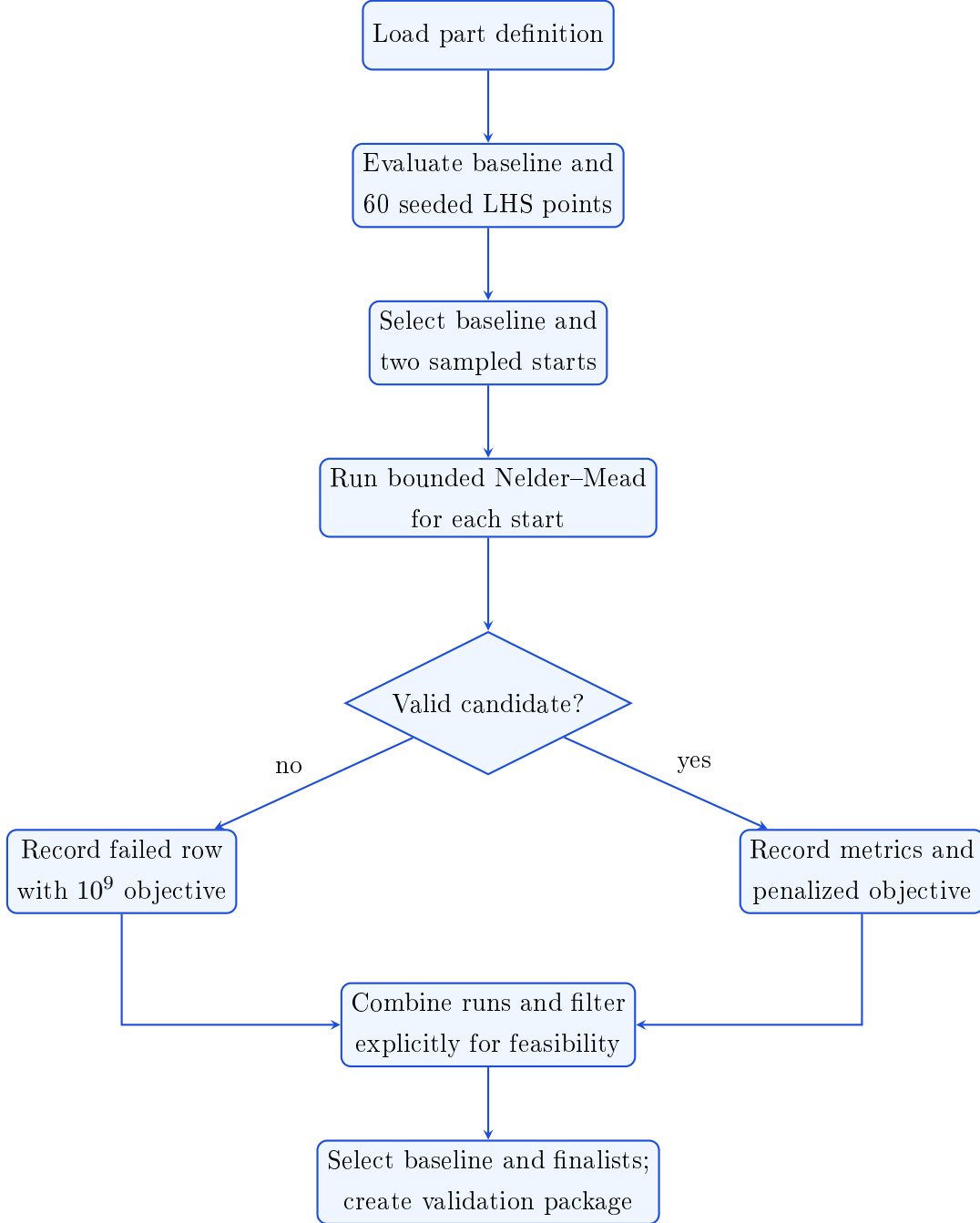


Figure 3.2: Sampling, optimization, and finalist-selection procedure

The selection step is deliberately separate from the optimizer’s minimum objective.

The validation package sorts only feasible rows by mass. It retains five finalists so that the lightest design does not become the only candidate for later FEA.

3.16 Run Logging and Dashboard

Each CSV log represents one part. The stable column order contains schema information, identifiers, mode, status, exact parameters, volume, mass, stress, factor of safety, displacement, objective, error, and timestamp. A companion `run.json` stores run-level metadata.

The Streamlit dashboard reads the records but does not change them. It can display convergence, parameter values, mass, constraints, and status. The dashboard is an evidence viewer, not a second source of calculation. This separation allows the report and dashboard to refer to the same stored rows.

3.17 Versioned Fusion Bridge

Schema version 2 requests contain:

- `schema_version`;
- `part_id`;
- `run_id`;
- `iteration`;
- `mode`;
- `parameters`; and
- `artifact_formats`.

Schema version 1 is retained for the original bracket proof of concept. It uses `parameters_mm` and implies the bracket.

The shared channel is `data/runs/active`. The external process takes an inter-process lock before it writes a request. It writes to a temporary file and then replaces the target atomically. NEMOBridge polls the request, validates the payload, rebuilds the specified part, writes artifacts, and returns a response. The external side accepts the response only when both `run_id` and `iteration` match.

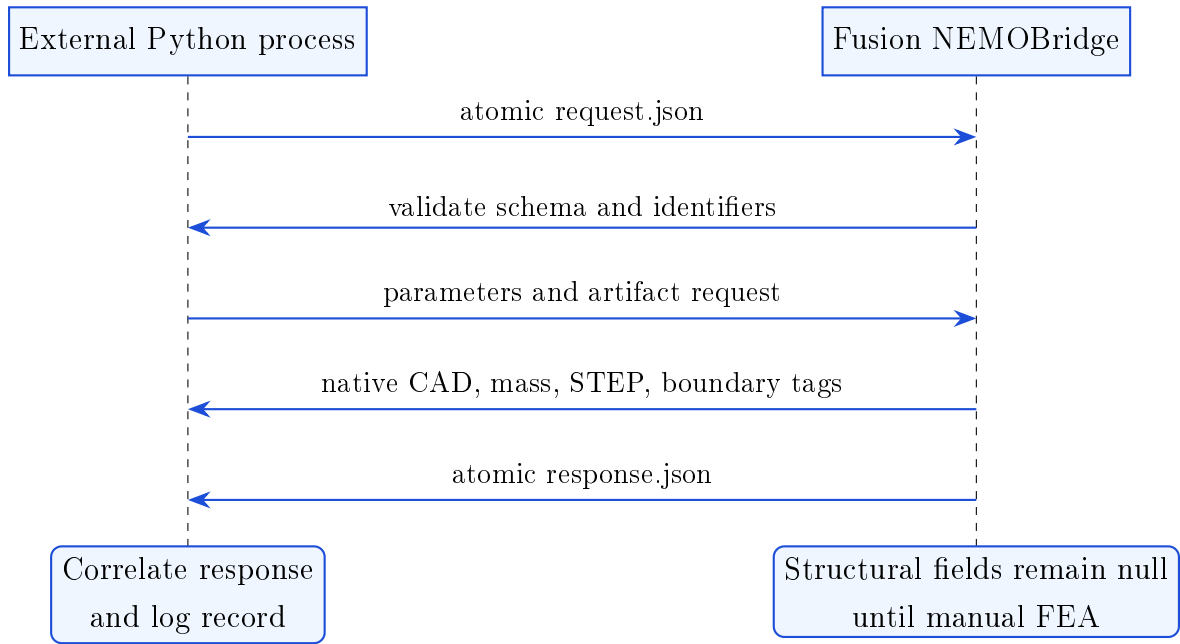


Figure 3.3: Serialized Fusion request and response sequence

The channel intentionally permits only one active CAD request. Concurrent requests would create a race over the same files. A future protocol can assign one directory per request or use a queue, but the current design favors a simple auditable sequence.

3.18 Native CAD Generation

The Fusion generator reads the same JSON definitions as the external package. It does not import `nemo` because Fusion uses a separate Python environment.

3.18.1 Bracket Generator

The bracket generator creates a rectangular baseplate, cuts four mounting holes, creates two triangular ribs, joins the bodies, and applies four long rib-root fillets. It checks that the result is one connected solid with positive volume. It identifies the mounting-hole cylindrical faces as the fixed-support role and the four upper sloping rib faces as the equipment-load role.

3.18.2 Padeye Generator

The padeye generator creates the doubler plate, tapered lug, circular crown, pin bore, two root fillets, and two overlapping gussets. It requires one connected solid. The underside of the doubler plate supplies the fixed-support role. The cylindrical pin-hole region supplies the pin-bearing role.

3.18.3 Stabilizer Generator

The stabilizer generator creates the root flange, lofted NACA outer envelope, root fillet, hollow interior, front and rear spars, three ribs, root insert, and pressure-band splits. It requires one connected solid. The root flange provides the fixed support. Four surface regions provide the pressure bands, and a tip region provides the displacement monitor.

Autodesk exposes physical-property and export functions through the public API [25, 26]. NEMO records mass and volume from the generated component and exports STEP for downstream exchange. STL is optional because it is a tessellated artifact rather than the preferred solid exchange format.

3.19 Semantic Boundary Metadata

For each required role, the generator stores one or more face signatures. Available fields include area, centroid, normal, bounding box, and cylindrical geometry. The generator fails the sweep acceptance check if a required role has no matching face.

This metadata has three functions. First, it documents what the generator believed the load or support region to be. Second, it helps the engineer find the same region in the manual study. Third, it makes topology problems visible during a sweep.

It does not automatically apply the FEA boundary condition. The engineer must still display the faces, compare them with the intended load path, and reject an incorrect selection.

3.20 CAD Generator Verification Campaign

Each generator was tested with its configured baseline and 20 seeded Latin-hypercube vectors. The acceptance criteria for each of the 21 cases were:

1. the request completed without an uncontrolled exception;
2. the result contained exactly one connected solid;
3. the physical volume was positive;
4. STEP export completed;
5. boundary metadata export completed; and
6. every required semantic boundary role matched at least one face.

The sweep does not solve FEA. It tests whether the generator and boundary contract remain usable across the bounded design space. It is intentionally separate from the offline test suite because it requires an open Fusion session and mutates the shared live channel.

3.21 Validation-Package Method

For each part, the package generator combines the baseline and five light feasible candidates. It writes:

- `validation_candidates.csv`;
- `validation_candidates.json`;
- one schema-version-2 Fusion request per candidate; and
- a `VALIDATION_CHECKLIST.md`.

The checklist records the analytical source row, parameters, predicted metrics, intended material, load and support roles, CAD status, CAD mass, FEA inputs, FEA outputs, convergence evidence, and engineering disposition. It prevents an unlabelled screenshot from becoming the only record of a result.

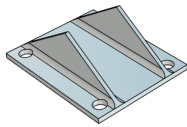
3.22 Bracket Proof-of-Concept Workflow

The bracket demonstration preserves the original project intent. A complete fresh-start flow is:

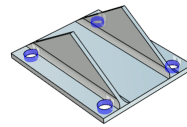
1. start Fusion and open the Scripts and Add-Ins dialog;
2. start NEMOBridge;
3. run the external bracket analytical pipeline;
4. inspect the sampled and optimized records;
5. generate the bracket validation package;
6. submit the baseline or one finalist request to NEMOBridge;
7. inspect the native bracket and exported artifacts;
8. open the Simulation workspace and create a Static Stress study;
9. assign the project-defined Aluminum 6061-T6 properties;
10. apply fixed constraints to the four mounting-hole cylindrical faces;
11. apply a total downward force of **1472 N** to the four upper sloping rib faces;
12. turn off *force per entity* so 1472 N is the combined load, not 1472 N on each face;
13. generate and inspect a parabolic mesh;
14. solve the study;

15. check reaction balance, stress, factor of safety, displacement, and result locations;
16. refine the mesh at holes and rib-root fillets;
17. compare the same monitored locations between meshes; and
18. record the results and evidence in the checklist.

The four upper sloping rib faces are the inclined top faces of the two ribs. There are two selectable top faces on each rib. They represent the equipment load interface.



(a) Bracket geometry in the Fusion report



(b) Four mounting-hole support regions

Figure 3.4: Bracket geometry and fixed-support setup extracted from the saved Fusion report. These images document study preparation. They do not validate the numerical result because the report used the wrong force magnitude.

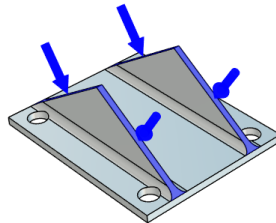


Figure 3.5: Downward load arrows on the four upper sloping rib faces. The image identifies the intended load region, but the saved report records 1.472 N instead of the required total integer load of 1472 N.

3.23 Manual Static Stress Quality-Control Procedure

The FEA procedure separates setup confirmation, numerical verification, and engineering interpretation.

3.23.1 Material Check

The material record must show density, Young's modulus, Poisson ratio, and yield strength. The values must match the candidate definition or a justified alternative. The engineer must not rely only on a library material name, because library revisions or custom data can differ.

3.23.2 Load Check

The engineer must confirm the magnitude, unit, direction, and distribution. For the bracket, the dialog must show 1472 N total. *Force per entity* must be off when the four faces share one total load. The arrow direction must match the intended downward equipment load.

3.23.3 Support and Contact Check

The four hole faces must be highlighted together. This idealization represents a rigid bolt-down and can create local peaks. A later detailed model should represent fasteners, bearing, contact, and preload when these effects affect the decision.

Every contact in the solid must be reviewed. Unintended separation or a disconnected body can invalidate the stiffness and load path.

3.23.4 Mesh Check

The record must include element order, global size, minimum size, growth or adjacency control, node count, and element count. Local refinement should target mounting holes, rib-root fillets, and load-introduction edges. At least three systematically refined meshes should be compared.

3.23.5 Equilibrium Check

The vector sum of the support reactions should oppose the applied force. For the corrected bracket case, the vertical reaction should be close to 1472 N. This check is simple and would have exposed the saved report's 1.472 N input.

3.23.6 Result Check

Maximum von Mises stress, minimum factor of safety, and maximum displacement must be recorded with their locations. A peak at a constrained edge can be a singularity. The engineer should also record stress at a stable nearby location and compare it across mesh refinements.

3.23.7 Acceptance Check

FEA acceptance requires the corrected load, matching material, intended boundaries, equilibrium, stable results, and the project constraints. Even after these checks, the study remains one load case. It does not establish fatigue, buckling, weld, bolt, corrosion, impact, or classification compliance.

3.24 Use of the Existing Fusion HTML Reports

The saved **Study Final.html** report identifies the study type, material, mesh settings, support faces, load faces, and result views. It records a parabolic mesh with 4781 nodes and 2355 elements, an average element size of 10% of the model scale, a 20% minimum element size, 10% adaptive refinement, and a 20% convergence tolerance.

The same report records a force magnitude of 1.472 N. The intended load is 1472 N. The numerical stress, factor of safety, displacement, and reactions from that report are invalid for the project load. They are not used in the optimization comparison or acceptance discussion.

The report images remain useful in three limited ways:

1. they show that the native bracket reached the Simulation workspace;
2. they show the intended support and load regions; and
3. they demonstrate why an audit must check inputs and reactions before accepting a solved study.

3.25 Extension to Padeye FEA

The padeye study will follow the same evidence sequence with part-specific boundaries and result regions. The underside of the doubler plate will provide the first fixed-support idealization. The 14 715 N design load will act through the pin-bearing region.

An initial linear screening model can use a distributed bearing load on the hole surface. A later detailed model should include a separate pin, clearance, contact, and any required nonlinear behavior. The review regions are the loaded hole, crown, net section, shear-out ligaments, lug root, gusset roots, doubler interface, and weld lines. Reaction balance must close to 14 715 N in the applied direction.

The final design assessment must also check the selected lifting code, load angle range, dynamic factors, material certification, weld class, fabrication tolerance, inspection, proof load, fatigue, and supporting structure.

3.26 Extension to Stabilizer FEA

The stabilizer study will fix the root-flange interface. The 11808 N resultant will be distributed over four pressure bands. The band forces are calculated by normalizing the weights 1.00, 0.93, 0.75, and 0.40. Pressure on each band equals its force divided by its actual CAD face area.

The study will monitor tip displacement and inspect the root fillet, front and rear spars, skin-spar intersections, rib junctions, insert termination, root flange, and trailing edge. Shell or solid element choice must represent the thin skins and local joints adequately. Buckling and fatigue require separate studies.

The prescribed load distribution is an interim structural envelope. A later hydrodynamic stage must define operational conditions and derive pressure from an accepted method. The structural model should then include positive and negative load cases, torsion, actuator loads, and relevant vessel motions.

3.27 Verification and Reproducibility

Offline verification uses focused tests during development and a full test suite before hand-off. The tests cover manifest loading, bounds, baseline feasibility, qualitative sensitivity, optimizer behavior, logging, bridge contracts, validation packages, dashboard reading, and generator contract helpers.

Fusion integration sweeps are deliberately opt-in because they require an open desktop application, take several minutes, and mutate the live channel. They do not perform FEA. Manual FEA evidence is recorded separately.

The repository records the exact finalist parameters and analytical values in CSV and JSON. Fusion responses record native CAD mass and artifact paths. The report asset script extracts Fusion report images and regenerates the plotted comparisons from these stored records. This arrangement allows later reviewers to trace a table or figure back to an inspectable source.

3.28 Methodological Safety Boundary

The values in this section are first-order screening estimates. They are not certified engineering results and do not establish manufacturing or service suitability.

The method was designed to reduce candidate count and improve traceability. It was not designed to certify marine hardware. A future engineering decision must use verified loads, applicable codes, detailed FEA, uncertainty assessment, fatigue and fabrication checks, physical testing where required, and qualified approval.

Chapter 4

Results and Discussion

4.1 Chapter Overview

This chapter reports software verification, analytical baselines, Latin-hypercube exploration, multi-start optimization, finalist selection, native CAD sweeps, CAD mass checks, and the bracket FEA quality-control finding. Analytical, CAD, and FEA results are reported separately. The chapter then answers the research questions and discusses the limitations of the evidence.

4.2 Software Verification Results

The recorded offline campaign passed 27 tests. Three Fusion-only tests were skipped because they require an open Fusion session and an active NEMOBridge. The offline tests covered:

- loading all registered part definitions;
- parameter bounds and exact units;
- baseline evaluation and feasibility;
- qualitative sensitivity to lighter and heavier geometry;
- random and Latin-hypercube sampling;
- bounded optimizer behavior;
- failed-evaluation containment;
- schema-version-1 and schema-version-2 bridge contracts;
- atomic request and correlated response handling;
- stable CSV logging;
- validation-package generation;

- dashboard data loading; and
- Fusion generator contract helpers.

The Fusion sweep remained a separate integration campaign. It requires desktop Fusion, changes `data/runs/active`, and can take several minutes per part. It does not solve FEA. This separation prevents a slow, state-dependent desktop test from being mistaken for an ordinary unit test.

4.3 Analytical Baselines

The values in this section are first-order screening estimates. They are not certified engineering results and do not establish manufacturing or service suitability.

Table 4.1 gives the configured baselines. All three passed the implemented factor-of-safety and displacement criteria.

Table 4.1: Analytical baseline screening results

Part	Mass (kg)	Stress (MPa)	FOS	Deflection (mm)	Volume (m ³)
Bracket	0.418125	41.5321	6.6455	0.195642	0.000154861
Padeye	8.847625	46.1070	5.9644	0.013429	0.001127086
Stabilizer	19.655593	86.9853	3.1730	3.336750	0.007279849

The bracket and padeye baselines had large yield-based margins. Their deflection estimates were also well below the project limits. The stabilizer had less factor-of-safety and displacement margin, although it remained analytically feasible. These differences affected the available mass reduction. A highly conservative baseline provides more room for a mass-only optimizer than a baseline already close to an active constraint.

4.4 Latin-Hypercube Exploration

Each part used 60 samples with seed 42. Table 4.2 summarizes the feasible counts and lightest feasible sampled designs.

Table 4.2: Seeded Latin-hypercube results

Part	Feasible samples	Lightest mass (kg)	FOS	Deflection (mm)
Bracket	55 of 60	0.245687	10.4945	0.073331
Padeye	60 of 60	5.923561	4.9324	0.010449
Stabilizer	47 of 60	16.482802	2.7674	4.107580

The sample stage achieved its three intended purposes. It exercised broad combinations of bounded variables, identified feasible low-mass starting points, and showed that the parts had different constraint activity.

The bracket had five infeasible sampled designs, but its lightest feasible sample still had high factor-of-safety margin. This indicates that broad random combinations did not automatically exploit the dimensions that control the screening stress.

All padeye samples were feasible. The result indicates that the declared bounds and simplified stress model produce a generous feasible region under the single project load case. It does not prove that every bounded padeye would pass pin contact, weld, fatigue, or governing lifting rules.

The stabilizer had 13 infeasible samples. Its lightest feasible sample was already close to both constraints compared with the other parts. The stabilizer therefore presented the clearest constrained search.

4.5 Multi-Start Optimization Results

Table 4.3 gives the lightest feasible row found in each local run. The start names identify the configured baseline and two sample-derived starts.

Table 4.3: Lightest feasible result from each Nelder–Mead start

Part	Start	Evaluations	Mass (kg)	FOS	Deflection (mm)
Bracket	Baseline	132	0.126545	2.5530	0.475840
	Sample 1	126	0.116928	2.5208	0.396650
	Sample 2	129	0.113396	2.5386	0.369832
Padeye	Baseline	108	3.837083	2.8528	0.026150
	Sample 1	116	3.609310	2.5767	0.026789
	Sample 2	109	3.547772	2.6224	0.027933
Stabilizer	Baseline	110	14.414066	2.5041	3.932243
	Sample 1	116	14.315361	2.5149	3.822300
	Sample 2	111	14.391670	2.5331	3.847508

Every part benefited from more than one start. The lightest bracket and padeye feasible designs came from the second sampled start. The lightest stabilizer feasible design came from the first sampled start. If the study had used only the configured baseline, the selected masses would have been higher for all three parts.

The bracket baseline-start result approached the 0.5 mm displacement limit, while the other two bracket results were governed more clearly by the factor-of-safety limit. The padeye displacement remained far from 0.5 mm in all runs, so its yield-based stress

controlled the search. The stabilizer operated close to the factor-of-safety limit while using approximately 76–79% of the permitted displacement.

4.6 Selected Analytical Finalists

The validation packages selected the lightest explicitly feasible candidates. Table 4.4 compares the selected finalist with its baseline.

Table 4.4: Selected analytical finalist compared with baseline

Part	Baseline (kg)	Finalist (kg)	Reduction	Final FOS	Defl. (mm)
Bracket	0.418125	0.113396	72.9%	2.5386	0.369832
Padeye	8.847625	3.547772	59.9%	2.6224	0.027933
Stabilizer	19.655593	14.315361	27.2%	2.5149	3.822300

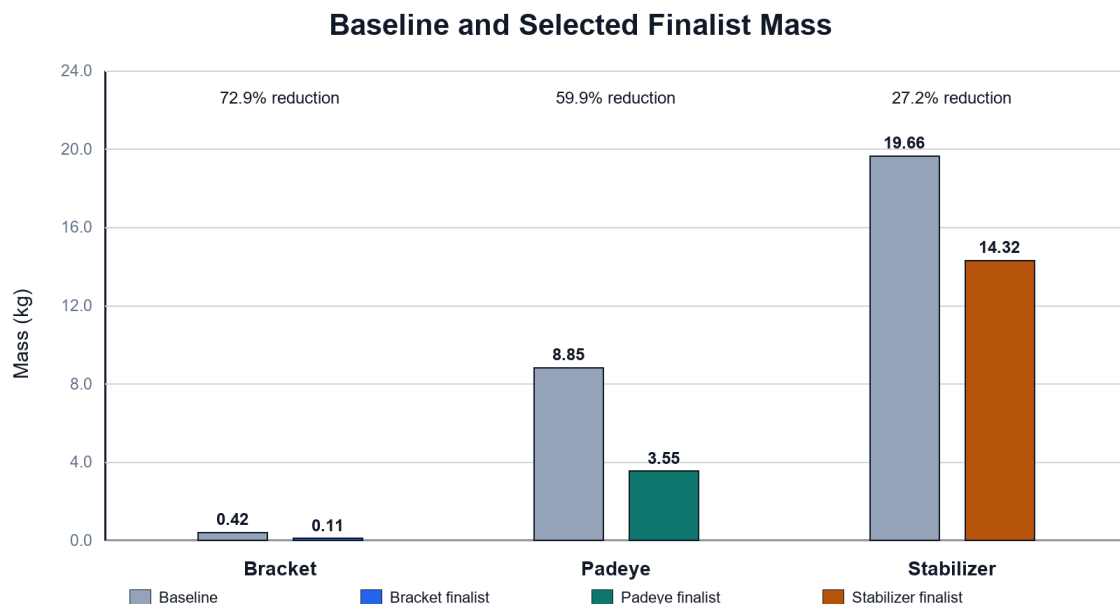


Figure 4.1: Analytical baseline and selected-finalist mass. Percentages are screening reductions within the declared parameterized searches.

The reductions are large, particularly for the bracket and padeye. They must be interpreted with the baseline and model scope. The baselines were configured reference designs, not independently weight-optimized industrial designs. The equations omit several failure modes. The percentage reduction therefore measures the opportunity found by the implemented screening model, not a validated service weight saving.

4.7 Finalist Constraint Use

Figure 4.2 normalizes the finalist responses. A factor-of-safety ratio of 1.0 means the finalist is exactly at the minimum factor of safety. A displacement ratio of 1.0 means it is exactly at the displacement limit.

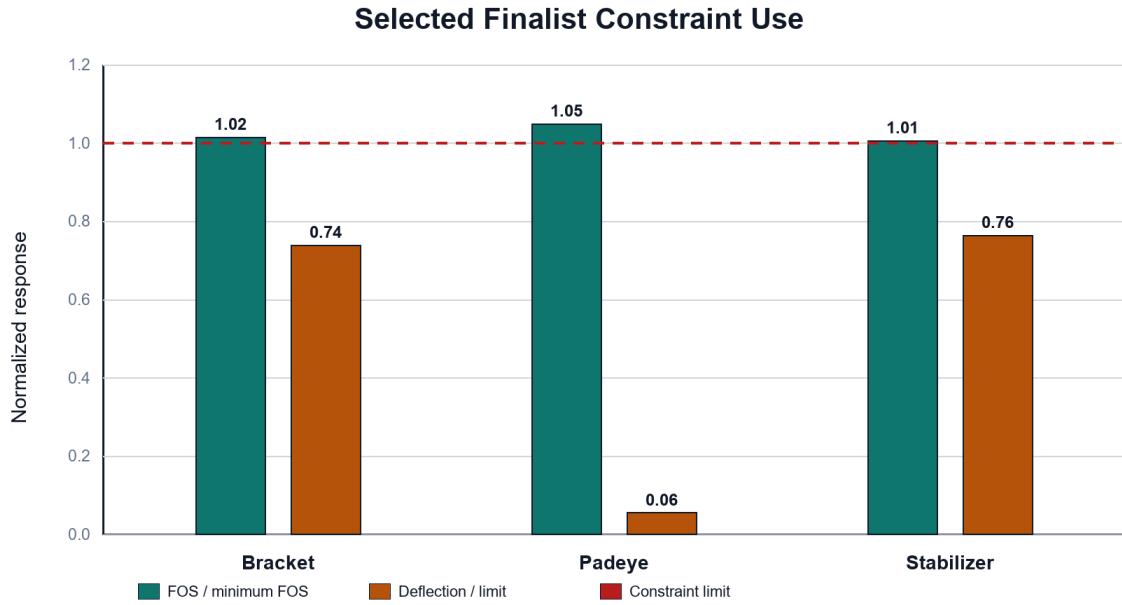


Figure 4.2: Normalized constraint use for the selected analytical finalists

All three finalists have factor-of-safety ratios only slightly above 1.0. This result is expected from a mass-only search with a yield constraint: the optimizer removes material until a constraint becomes active. The result also shows why screening finalists need conservative backup candidates. A small change in load, material, stress concentration, or model bias could make the lightest row infeasible.

The bracket displacement ratio is approximately 0.74. The padeye displacement ratio is approximately 0.056, so the simplified padeye search is almost entirely controlled by stress. The stabilizer displacement ratio is approximately 0.76. A later FEA model can change which constraint governs.

4.8 Parameter Activity and Engineering Interpretation

4.8.1 Bracket

The selected bracket used approximately 100.03 mm baseplate length, 80.94 mm width, 4.04 mm baseplate thickness, 33.26 mm rib height, 3.00 mm rib thickness, and 3.53 mm fillet radius. The length, width, plate thickness, and rib thickness were near their lower bounds.

The pattern reflects the analytical equations. Plan dimensions and thickness reduce volume directly. Rib height retains section depth and bending resistance with less mass than a uniformly thick plate. This is a plausible broad trend, but the candidate has little allowance for corrosion, fabrication tolerance, weld size, bearing, or local plate action. The lower-bound activity is therefore a prompt to review the bounds and omitted constraints, not proof that the minimum gauge is suitable.

4.8.2 Padeye

The selected padeye used approximately 220.39 mm base length, 150.47 mm base width, 8.05 mm base thickness, 141.73 mm lug height, 12.16 mm lug thickness, 80.00 mm neck width, 70.00 mm gusset height, 6.21 mm gusset thickness, and 25.00 mm fillet radius.

Many thickness and envelope variables approached their lower bounds, while the fillet radius reached its upper bound. In the analytical model, a large fillet reduces the assumed concentration factor with a relatively modest volume cost. The optimizer therefore spends material on the fillet while reducing other dimensions. A three-dimensional contact and weld model may change this balance.

4.8.3 Stabilizer

The selected stabilizer used approximately 4.01 mm skin, 25.0% front-spar position, 72.0% rear-spar position, 7.85 mm front spar, 6.53 mm rear spar, 5.70 mm ribs, 221.48 mm root insert, 13.66 mm insert thickness, and 18.33 mm root fillet.

The skin approached its lower bound, while the spars moved apart. A larger spar separation can increase effective section depth. The optimizer retained root reinforcement to protect the factor-of-safety limit. This response is consistent with the root-section model. It remains sensitive to the assumed pressure distribution and omitted torsion and buckling.

4.9 Penalty Objective and Explicit Feasibility

The smallest penalized objective in each stabilizer run was infeasible:

Table 4.5: Infeasible minimum-objective rows from the stabilizer searches

Start	Objective	Mass (kg)	FOS
Baseline	13.9702	13.74695	2.38188
Sample 1	13.7477	13.61934	2.41042
Sample 2	13.7335	13.61934	2.41554

The penalty increased the objective, but the low mass remained attractive enough

to produce a smaller value than the feasible rows. If the package had selected the minimum objective without a feasibility filter, it would have promoted a failed design. The implemented selection avoided this error.

This result shows a limitation and a strength. The fixed penalty weight is not strong enough to make every run’s objective minimum feasible. However, the system does not treat the penalty as the acceptance rule. Explicit feasibility is the acceptance rule. Future work can compare adaptive penalties, feasibility-first ranking, or a constrained derivative-free method.

4.10 Native CAD Generator Sweeps

The final sweeps used generator version 2.4. Table 4.6 reports their status.

Table 4.6: Baseline plus 20-vector native CAD sweep results

Part	Artifacts created	Integration passes	Recorded observation
Bracket	21 of 21	21 of 21	Completed in approximately 238.19 s.
Padeye	21 of 21	21 of 21	Completed in approximately 157.19 s.
Stabilizer	21 of 21	Final read timed out	All 21 artifacts were created. The outer test reached its 30-minute timeout while reading the final response; direct artifact audit confirmed generation.

Every audited case contained one connected solid, positive volume, STEP output, boundary metadata, and non-empty required boundary roles. The result supports generator behavior over the seeded bounded vectors. It does not prove behavior for every possible point in the continuous design space.

The stabilizer timeout is reported rather than hidden. The evidence supports successful artifact generation for all 21 vectors, but it does not support a claim that the complete outer integration test exited normally. The event also shows why long-running

desktop automation should preserve intermediate artifacts and per-case records.

4.11 Analytical and Native-CAD Mass Comparison

Table 4.7 compares analytical and Fusion CAD mass for each baseline and selected finalist. The percentage difference is calculated relative to the analytical mass.

Table 4.7: Analytical and native-CAD mass comparison

Part	Design	Analytical (kg)	Fusion CAD (kg)	Difference
Bracket	Baseline	0.418125	0.422813	+1.12%
Bracket	Finalist	0.113396	0.114705	+1.15%
Padeye	Baseline	8.847625	8.897275	+0.56%
Padeye	Finalist	3.547772	3.682599	+3.80%
Stabilizer	Baseline	19.655593	19.297290	-1.82%
Stabilizer	Finalist	14.315361	14.400328	+0.59%

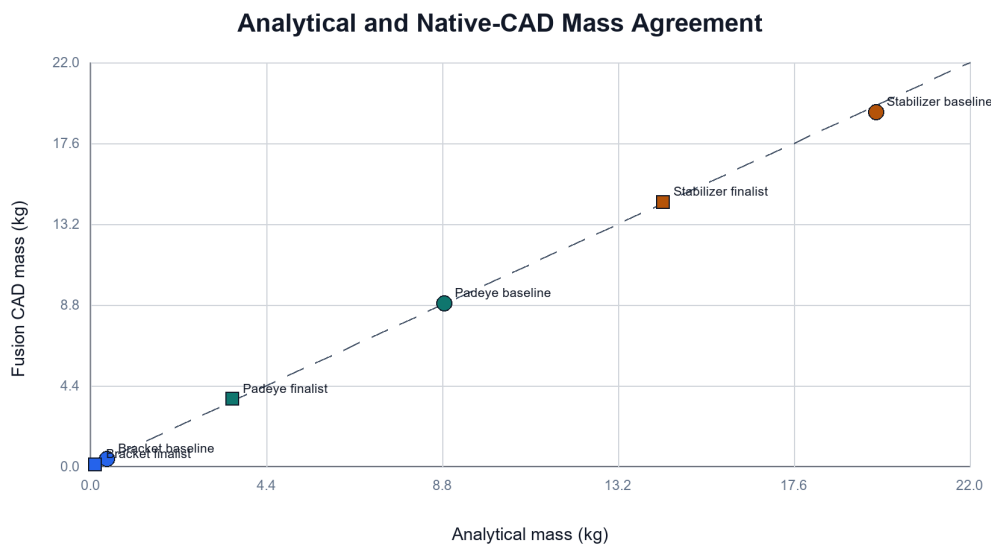


Figure 4.3: Analytical mass compared with Fusion native-CAD mass. The diagonal shows exact agreement.

All six differences were within 3.8%. The agreement supports the broad volume approximations used for mass ranking. The padeye finalist showed the largest difference, which is consistent with greater geometric complexity around the lug crown, fillets, and gusset intersections.

The agreement does not validate stress. Mass depends primarily on volume and density. Structural response depends on the load path, section properties, local geometry, support idealization, contact, and mesh. The CAD response therefore remains correctly marked as partial.

4.12 Bracket FEA Pilot and Quality-Control Finding

The saved Fusion report documents a Static Stress study with the bracket geometry, Aluminum 6061-T6 properties, four mounting-hole support faces, four upper rib load faces, and a parabolic mesh. The report also contains result views. However, the load table records 1.472 N.

The intended manual input was the integer value 1472 N. The saved input is one-thousandth of the required force. The report's very small stress and displacement and very high factor of safety are therefore not structural results for the project load.

This finding changes the result status:

- the geometry and interface images are valid setup evidence;
- the support and load face locations are useful evidence;
- the mesh metadata documents one solver configuration;
- the numerical stress, factor of safety, displacement, and reaction are rejected; and
- a corrected 1472 N, reaction-balanced, mesh-converged study remains required.

The pilot was still valuable. It showed that successful solver completion is not sufficient evidence. The workflow must compare the intended load with the stored load and compare the applied force with support reactions.

4.13 Comparison with the Project Objectives

Table 4.8: Achievement of the specific objectives

No.	Objective outcome	Status
1	Unit-aware definitions for three parts were implemented.	Achieved.
2	First-order mass, stress, FOS, and displacement models were implemented.	Achieved as screening models.
3	Each part completed 60 seeded LHS evaluations.	Achieved.
4	Each part completed three bounded scaled Nelder–Mead starts.	Achieved.
5	Invalid and unavailable evaluations return controlled failures.	Achieved.
6	A versioned atomic file bridge connects external Python and Fusion.	Achieved for serialized requests.

No.	Objective outcome	Status
7	Native Fusion CAD, physical properties, STEP, and boundary metadata were generated.	Achieved.
8	Baseline plus 20-vector sweeps were executed for all parts.	Achieved, with the stabilizer outer-process timeout disclosed.
9	Baseline and finalist validation packages were generated.	Achieved.
10	The bracket completed the workflow through manual FEA preparation.	Partly achieved; the stored FEA load was wrong and must be corrected.
11	The padeye and stabilizer use the shared architecture.	Achieved through analytical, CAD, and validation preparation.
12	Remaining engineering validation work is stated.	Achieved.

4.14 Answers to the Research Questions

4.14.1 Part-Aware Architecture

The registry, evaluator, sampler, optimizer, logger, bridge request, validation package, and dashboard processed all three part identifiers. Part-specific analytical equations and CAD generators were dispatched through shared interfaces. This evidence supports the first research question. It does not imply that every future part can be added without new part-specific code.

4.14.2 Lower-Mass Feasible Candidates

The sampling and multi-start searches found lower-mass analytically feasible candidates for all three parts. The selected screening reductions were 72.9%, 59.9%, and 27.2%. The evidence supports “best feasible design found within the parameterized search.” It does not support global optimality or validated service savings.

4.14.3 Native CAD and Boundary Roles

The Fusion generator created all seeded artifacts and non-empty required boundary roles. The bracket and padeye completed all 21 integration cases. The stabilizer artifact set

completed, although the external test timed out while reading the final response. The evidence supports generator and boundary-contract operation over the sampled points.

4.14.4 Analytical and CAD Mass Agreement

Analytical and CAD mass differed by no more than 3.8% for the six baseline and finalist comparisons. This supports the volume models as practical mass screening estimates. It does not validate structural response.

4.14.5 Evidence Required Before Design Acceptance

All finalists require corrected and mesh-converged FEA, applicable design-code checks, verified loads and material properties, local failure-mode assessment, fatigue, buckling where relevant, corrosion and fabrication allowances, physical testing where required, and qualified engineering review.

4.14.6 Lessons from the Bracket Proof of Concept

The bracket showed that the file bridge and native CAD can support a clear human-reviewed handoff. It also showed that a decimal-point or unit error can invalidate an apparently complete FEA report. Padeye and stabilizer work must therefore use integer-compatible force inputs where required, input checklists, reaction balance, and recorded convergence evidence.

4.15 Limitations of the Results

The analytical models are intentionally first order. They use simplified load paths, nominal section properties, and approximate concentration factors. The reported constraints include yield-based factor of safety and displacement only.

The search used a finite sample and three local starts. A different seed, start, penalty, bound, or model can produce a different candidate. Several finalists lie near lower bounds, so the selected bounds strongly influence the result.

The CAD sweeps verify sampled geometry, not the full continuous design space. They also do not check manufacturing drawings, tolerances, weld access, or assembly.

The manual FEA evidence is incomplete. The available bracket report has an incorrect load and cannot validate stress or displacement. Padeye and stabilizer solver campaigns remain future work. No physical tests were performed.

4.16 Engineering Interpretation

The results support the central workflow claim: inexpensive analytical evaluation can reduce a large design set to a small group of traceable candidates, and native CAD can check geometry and mass before manual FEA. They do not support certification or fabrication.

The most important result is not the largest percentage reduction. It is the separation of claims. The system recorded analytical predictions, returned partial CAD responses without solver values, disclosed the stabilizer timeout, and rejected a completed Fusion report after finding the wrong load. This behavior is consistent with an auditable engineering workflow.

Chapter 5

Project Workplan and Budget

5.1 Chapter Overview

This chapter records the zero direct-cash budget and the project schedule from September 2026 to March 2027. The schedule is intentionally concise. Detailed technical procedures and verification activities are described in Chapters 3 and 4.

5.2 Zero Direct-Cash Budget

The project had a direct-cash budget of **KES 0**. No equipment, commercial software, cloud compute, data, printing, travel, fabrication, or testing service was purchased from project funds.

Table 5.1: Direct-cash project budget

Category	Basis for zero direct expenditure	Budget (KES)
Development hardware	Existing personal or university computers were used. No computer or upgrade was purchased for the project.	0
CAD and simulation software	Autodesk educational access was used. No commercial license was purchased from project funds.	0
Python and report software	Python, project libraries, Git, and the LaTeX toolchain were available without direct project charges.	0
Cloud computing and storage	All computation and working data remained local or used existing remote access. No paid cloud service was purchased.	0
Internet and electricity	Existing personal or university services supported the work. No separate project subscription or utility purchase was budgeted.	0
Printing, travel, and communication	The report was prepared for digital submission. No project-specific printing, travel, or communication expense was charged.	0
Prototype fabrication and testing	No physical component or paid laboratory test was included in the completed scope.	0
Contingency	The contingency consisted of schedule margin and scope control, not cash.	0
Total direct-cash budget		0

The zero budget does not mean that the project used no resources. Existing computers, Autodesk educational access, internet, electricity, supervision, and student labor were in-kind inputs. Table 5.2 records them without assigning invented monetary values.

Table 5.2: In-kind resources used but not monetized

Resource	Provision basis	Project use
Computers	Existing personal or university access	Python development, tests, Fusion CAD, simulation, report compilation, and demonstration.
Autodesk Fusion	Educational entitlement	Native CAD generation, physical properties, STEP export, and manual Static Stress preparation.
Open-source software	Public software licenses	Optimization, testing, data records, dashboard, charts, version control, and report production.
Internet and electricity	Existing personal or institutional access	Research, documentation, source synchronization, and software operation.
Student labor and supervision	Academic project activity	Engineering definition, implementation, review, testing, analysis, documentation, and presentation preparation.

5.3 Budget Control

The architecture helped keep the direct cost at zero. Analytical equations handled the large search, while manual FEA was reserved for a few candidates. The project used the in-application Fusion API and local JSON files instead of paid cloud automation. Open-source Python tools and digital reporting avoided software and printing costs.

Prototype fabrication and paid testing were not treated as unnecessary. They were placed outside the completed academic scope. A service-ready design would need a separate budget for material certification, fabrication, inspection, instrumentation, proof loading, and any required hydrodynamic testing.

5.4 Workplan: September 2026 to March 2027

The project was organized over seven months. September and October established the research basis, engineering definitions, and bracket proof of concept. November and December developed the search, records, and multi-part architecture. January and Febru-

ary focused on native CAD generation, verification, validation preparation, and results. March was reserved for the final report, demonstration, corrections, and submission.

The Fusion proof of concept was the main technical decision point. When the public API did not provide dependable automated Static Stress control, the project retained analytical optimization and native CAD, then changed the solver step to a manual, checklist-based process. Figure 5.1 summarizes this sequence. Documentation and report writing overlap the implementation work because assumptions and evidence were recorded throughout.

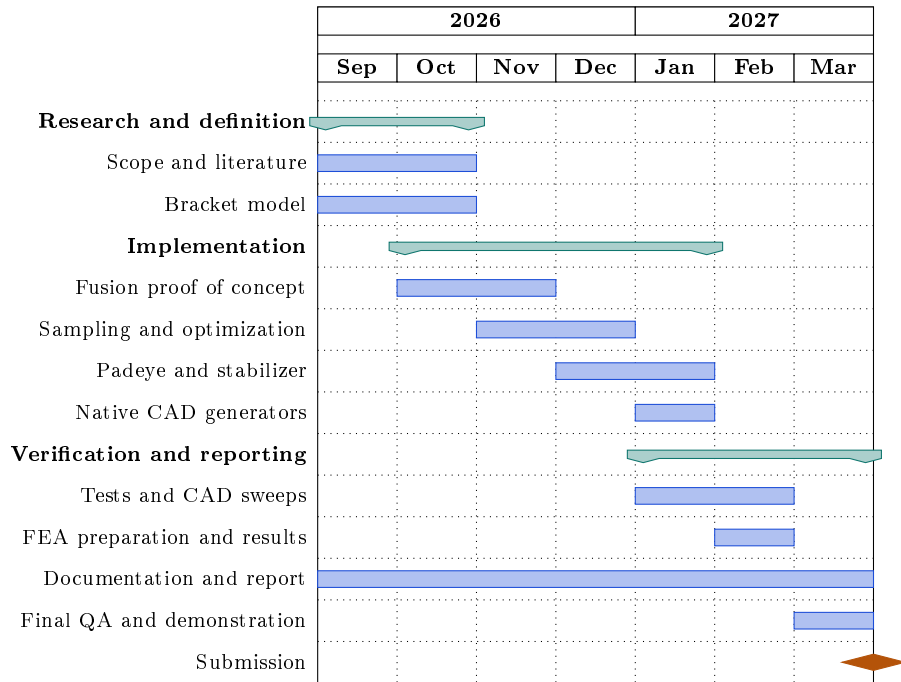


Figure 5.1: Portrait Gantt chart for September 2026 to March 2027

Chapter 6

Conclusions and Recommendations

6.1 Chapter Overview

This chapter states the conclusions, answers the research questions, identifies the project contributions, and gives recommendations for validation and future development. The conclusions remain limited to the evidence reported in the previous chapters.

6.2 Overall Conclusion

The project developed and evaluated an auditable, resource-conscious workflow for the parameterized mass optimization of three marine structural component archetypes. The completed system connects unit-aware engineering definitions, first-order screening, seeded design-space exploration, bounded multi-start Nelder–Mead search, failure-contained evaluation, traceable run records, native Fusion CAD, semantic boundary metadata, validation packages, and a results dashboard.

The workflow found lower-mass analytically feasible candidates for the bracket, padeye, and stabilizer. The selected screening reductions were 72.9%, 59.9%, and 27.2% relative to the configured baselines. Native Fusion CAD mass remained within 3.8% of the analytical mass for every baseline and selected-finalist comparison. These results support the use of the analytical volume models for early mass ranking and the use of the Fusion bridge for geometry and artifact verification.

The results do not establish structural qualification. The analytical models omit important local and service failure modes. The bracket FEA report used 1.472 N instead of the required integer input of 1472 N. Its solver values were therefore rejected. Padeye and stabilizer manual FEA campaigns remain future work. The finalists are screening candidates that require corrected, mesh-converged analysis, governing design checks, verified loads and materials, and qualified review.

The central project outcome is therefore not a claim that three components are ready for manufacture. It is a traceable reference process that reduces a large candidate

set, generates checkable CAD, exposes missing solver evidence, and prepares a small candidate set for disciplined engineering validation.

6.3 Answers to the Research Questions

6.3.1 Can One Architecture Support Three Part Types?

Yes, within the implemented scope. The registry, sampler, optimizer, logger, bridge schema, validation-package generator, and dashboard processed all three part identifiers. Each part supplied its own manifest, analytical evaluator, native CAD generator, and boundary selectors through shared interfaces. This result shows part-aware extensibility. It does not mean a fourth component can be added without new engineering and generator code.

6.3.2 Did Sampling and Multi-Start Search Find Lighter Feasible Designs?

Yes, according to the implemented screening models. Every part completed 60 seeded Latin-hypercube evaluations and three local searches. Every selected finalist passed the implemented factor-of-safety and displacement constraints. Multiple starts improved the best feasible mass for all three parts compared with the baseline start alone.

The result is not a global-optimum proof. The sample is finite, Nelder–Mead is local, the penalties are fixed, and the bounds define the available design family. The correct conclusion is that the method found lower-mass feasible candidates within the declared parameterized searches.

6.3.3 Did Native CAD and Boundary Identification Remain Usable?

The bracket and padeye completed all 21 baseline-plus-sample integration cases. The stabilizer created all 21 artifacts, although the outer test process timed out while reading the final response. Direct artifact audit confirmed one solid, positive volume, STEP export, and required boundary metadata for the generated cases.

This evidence supports the generator contract over the tested vectors. It does not prove the absence of every possible topology failure in the continuous design space. It also does not validate the FEA boundary idealization. Human inspection remains necessary.

6.3.4 How Closely Did Analytical and CAD Mass Agree?

The six baseline and selected-finalist comparisons differed by 0.56% to 3.80% in magnitude. The padeye finalist showed the largest difference. The close agreement supports

the analytical volume models for mass screening.

Mass agreement must not be extended to stress agreement. The CAD response does not contain structural solver results. Local stress and displacement depend on boundary conditions, contact, mesh, and three-dimensional load paths that are not established by volume agreement.

6.3.5 What Evidence Is Still Required?

The next evidence stage must include:

1. verified material and load inputs;
2. realistic supports, fasteners, contacts, and load introduction;
3. mesh-convergence evidence at stable monitored locations;
4. reaction-force balance;
5. detailed stress, displacement, and factor-of-safety results;
6. governing failure-mode checks;
7. applicable marine or lifting-code assessment;
8. fatigue, buckling, corrosion, fabrication, tolerance, and inspection checks as relevant;
9. uncertainty and sensitivity assessment; and
10. proof load or other physical validation where required.

6.3.6 What Did the Bracket Proof of Concept Teach?

The bracket demonstrated the intended human-reviewed chain from a parameterized definition to a selected design, native CAD, semantic support and load regions, and manual FEA preparation. It also exposed a serious quality-control issue. The saved study solved successfully with a force that was one-thousandth of the intended value.

The principal lesson is that solver completion is not acceptance. A study checklist must compare the intended and stored load, confirm force-per-entity behavior, inspect supports, and compare total reactions with the applied force. The same checks must precede interpretation of future padeye and stabilizer contours.

6.4 Contribution of the Work

6.4.1 Engineering-Definition Contribution

The project centralizes parameters, units, bounds, baselines, materials, loads, constraints, fixed geometry, and boundary roles. This design reduces cross-layer inconsistency and gives later work a traceable source for each engineering input.

6.4.2 Optimization-Implementation Contribution

The project combines scaled coordinates, bound clipping, seeded LHS, three Nelder–Mead starts, caching, failed-evaluation containment, stable logging, and explicit feasible filtering. The stabilizer result confirms the importance of feasible filtering when a penalty minimum remains infeasible.

6.4.3 CAD-Handoff Contribution

The project provides a versioned and inspectable exchange between an external Python process and a Fusion add-in. Native CAD, physical properties, STEP, and semantic boundary metadata are returned without mislabeling CAD output as FEA.

6.4.4 Evidence-Management Contribution

The report separates software, analytical, CAD, FEA, physical-validation, and qualification evidence. It records partial status when only CAD evidence exists. It also retains and explains a failed FEA input case. This separation improves auditability and academic credibility.

6.4.5 Practical and Educational Contribution

The workflow was completed with a KES 0 direct-cash budget by using existing hardware, educational Fusion access, open-source Python tools, local computation, and digital documentation. The result provides a reusable teaching example for model assumptions, optimization claims, desktop CAD automation, and simulation quality control.

6.5 Achievement of the Aim

The aim was achieved at the workflow-development and validation-preparation level. The project produced a working multi-part software artifact, analytical search results, native CAD evidence, boundary metadata, validation packages, and a preserved bracket proof of concept.

The aim was not achieved at the level of certified structural design. The project did not complete corrected mesh-converged FEA for the three selected finalists, physical tests, or governing-code qualification. These activities were outside the achieved evidence level and are explicitly recommended.

6.6 Recommendations

6.6.1 Immediate Bracket Validation

1. Recreate the bracket Static Stress study with a total downward **1472 N** force on the four upper sloping rib faces.
2. Turn off force-per-entity so the four faces share the total force.
3. Confirm the four mounting-hole cylindrical faces are the only fixed supports in the first idealization.
4. Record a total vertical support reaction close to 1472 N.
5. Use at least three systematically refined parabolic meshes.
6. Record stress at the peak and at stable nearby monitoring regions.
7. Compare the baseline, lightest finalist, and at least two conservative finalists.
8. Add bolt bearing, fastener, weld, local plate, fatigue, corrosion, and fabrication checks before selecting a design.

6.6.2 Padeye Validation

1. Confirm the governing lifting standard and load combinations before final sizing.
2. Apply 14 715 N through a verified bearing region for the first screening study.
3. Develop a later pin-contact model with clearance and nonlinear contact when the decision requires it.
4. Check net section, bearing, shear-out, lug root, gussets, welds, doubler plate, and supporting structure.
5. Include out-of-plane load, load-angle variation, fatigue, fabrication, inspection, and proof-load requirements.

6.6.3 Stabilizer Validation

1. Replace the assumed pressure distribution with a load case derived from an accepted hydrodynamic method, CFD, model tests, or measured data.

2. Apply positive, negative, and torsional load cases as relevant.
3. Check root stress, tip displacement, spar and rib junctions, skin–spar load transfer, local panel buckling, insert termination, flange, and trailing edge.
4. Include fatigue, corrosion, actuator interaction, vibration, and hydrodynamic uncertainty.
5. Select shell, solid, or mixed elements based on a convergence and idealization study.

6.6.4 Software Development

1. Implement the documented `open_fea` contract with a real mesh and solver backend before exposing that mode as available.
2. Add adaptive or feasibility-first constraint handling and compare it with the fixed penalty.
3. Add sensitivity and uncertainty analysis for loads, material properties, concentration factors, and model coefficients.
4. Add one request directory per CAD job or a queue before allowing concurrency.
5. Record software version, Fusion version, generator version, and input hashes in every validation package.
6. Add automatic comparison between the requested load and any imported solver report when a dependable report parser is available.

6.6.5 Model Calibration

After corrected FEA results exist, compare analytical and FEA response at the same candidate and load. If a systematic bias appears, revise the analytical coefficients or fit a correction model. Calibration must use separate candidates for fitting and checking. A corrected model should not be declared valid outside the parameter and load range supported by its evidence.

6.7 Recommended Validation Sequence

The next project phase should follow this sequence:

1. freeze the baseline and candidate parameter records;
2. verify material and load pedigree;
3. complete the corrected bracket FEA campaign;

4. compare analytical and FEA trends;
5. revise bounds or equations if the bracket model is non-conservative;
6. complete padeye linear screening FEA;
7. complete padeye pin-contact and code checks;
8. derive the stabilizer hydrodynamic load cases;
9. complete stabilizer static, buckling, and fatigue studies;
10. select conservative finalists rather than only the lightest rows;
11. fabricate and proof-test representative hardware where required; and
12. obtain qualified engineering review before service use.

6.8 Future Research

One future direction is a formal multi-fidelity optimizer. Selected FEA results could correct analytical predictions or train a surrogate with uncertainty. The optimization strategy could then select new high-fidelity points where they provide the greatest decision value.

A second direction is reliability-based design. Loads, material properties, fabrication dimensions, corrosion loss, and model error can be represented as uncertain quantities. Candidate selection could then use probability of failure or reliability index instead of one deterministic factor of safety.

A third direction is multi-objective optimization. Mass can be balanced against fabrication cost, fatigue damage, displacement, or carbon impact. The result would be a Pareto set rather than one mass minimum.

A fourth direction is manufacturing-aware optimization. Minimum gauges, stock sizes, tool access, bend radii, weld size, inspection access, additive manufacturing constraints, and tolerance sensitivity can become explicit variables or constraints.

A fifth direction is broader marine integration. The stabilizer can connect to CFD or measured pressure data. The padeye can connect to lifting-operation load cases. The bracket can connect to equipment vibration and foundation stiffness. Each extension must preserve the evidence and claim boundaries used in the current project.

6.9 Final Statement

NEMO shows that a student-scale engineering workflow can automate the repeatable parts of parametric search and CAD generation while leaving high-consequence structural judgment visible. The strongest conclusion is not that the lightest analytical candidates

are ready for use. It is that the workflow makes it difficult to confuse a prediction, a CAD property, a solver setup, and a validated design.

The project should therefore be continued as a validation program. The current repository provides the definitions, candidate history, CAD generators, boundary roles, checklists, and proof-of-concept lessons needed for that work.

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Appendix A

Reproduction and Validation Material

A.1 Fresh-Start Command-Prompt Procedure

The following procedure uses Windows Command Prompt. It does not require PowerShell.

1. Open Command Prompt.
2. Change to the repository:

```
1 cd /d C:\Users\muigu\Documents\Projects\NEMO
```

Listing A.1: Open the NEMO repository

3. Create or update the project environment:

```
1 build.bat setup
```

Listing A.2: Set up the project

4. Run the offline checks:

```
1 build.bat check
```

Listing A.3: Run offline verification

5. List the registered parts:

```
1 .env\Scripts\nemo.exe parts
```

Listing A.4: List the registered components

6. Evaluate the three baselines:

```
1 .env\Scripts\nemo.exe evaluate --part bracket
2 .env\Scripts\nemo.exe evaluate --part padeye
3 .env\Scripts\nemo.exe evaluate --part stabilizer
```

Listing A.5: Evaluate the analytical baselines

A.2 Bracket Proof-of-Concept Procedure

1. Start Autodesk Fusion.
2. Open *Utilities*, then *Scripts and Add-Ins*.
3. Select and start NEMOBridge.
4. If a bracket opens automatically, leave it available. NEMOBridge can load or create the demonstration design during startup.
5. Open Command Prompt and change to the repository.
6. Run:

```
1 cd /d C:\Users\muigu\Documents\Projects\NEMO
2 .venv\Scripts\python.exe demos\bracket-proof-of-concept\manual_fusion_handshake.
  py
```

Listing A.6: Run the preserved bracket proof of concept

7. Return to Fusion and inspect the regenerated bracket.
8. For the complete manual study, follow Section 3.22.

A.3 Schema-Version-2 Request Example

```
1 {
2   "schema_version": 2,
3   "part_id": "bracket",
4   "run_id": "demo_bracket",
5   "iteration": 0,
6   "mode": "fusion_cad",
7   "parameters": {
8     "baseplate_length": 150.0,
9     "baseplate_width": 100.0,
10    "baseplate_thickness": 8.0,
11    "rib_height": 40.0,
12    "rib_thickness": 6.0,
13    "fillet_radius": 5.0
14  },
15   "artifact_formats": ["step", "boundary_tags"]
16 }
```

Listing A.7: Illustrative bracket CAD request

A.4 CAD-Only Response Example

```
1 {
2   "schema_version": 2,
3   "part_id": "bracket",
4   "run_id": "demo_bracket",
5   "iteration": 0,
6   "status": "partial",
7   "metrics": {
8     "mass_kg": 0.4228127502,
9     "volume_m3": 0.0001565973,
10    "max_stress_mpa": null,
11    "factor_of_safety": null,
12    "max_deflection_mm": null,
13    "objective_value": null
14  },
15  "artifacts": {
16    "step": "...\\bracket_0000.step",
17    "boundary_tags": "...\\bracket_0000.boundaries.json"
18  },
19  "error": "Native CAD generation and mass extraction completed.
20           Static-stress results require manual Fusion validation."
21 }
```

Listing A.8: Illustrative partial CAD response

The partial status is intentional. It prevents a CAD success from being interpreted as an FEA success.

A.5 Selected Finalist Parameters

Table A.1: Selected bracket finalist parameters

Parameter	Value (mm)
Baseplate length	100.0328
Baseplate width	80.9380
Baseplate thickness	4.0429
Rib height	33.2576
Rib thickness	3.0000
Fillet radius	3.5308

Table A.2: Selected padeye finalist parameters

Parameter	Value (mm)
Base length	220.3878
Base width	150.4654
Base thickness	8.0484
Lug height	141.7256
Lug thickness	12.1613
Neck width	80.0000
Gusset height	70.0000
Gusset thickness	6.2101
Fillet radius	25.0000

Table A.3: Selected stabilizer finalist parameters

Parameter	Unit	Value
Skin thickness	mm	4.0106
Front spar position	% chord	25.0000
Rear spar position	% chord	72.0000
Front spar thickness	mm	7.8522
Rear spar thickness	mm	6.5297
Rib thickness	mm	5.6971
Root insert length	mm	221.4762
Root insert thickness	mm	13.6624
Root fillet radius	mm	18.3273

A.6 Manual FEA Validation Record

Each baseline and finalist FEA record should contain the fields in Table A.4.

Table A.4: Minimum manual FEA validation checklist

Record item	Required evidence
Candidate identity	Part identifier, candidate identifier, source run, iteration, parameter table, schema version, and generator version.
Geometry	Native CAD screenshot, connected-body count, volume, mass, and STEP path.
Material	Name, density, Young's modulus, Poisson ratio, yield strength, and source.

Record item	Required evidence
Load	Magnitude, unit, derivation, direction, selected faces, distribution, and force-per-entity state.
Support	Constraint type, selected faces, physical interpretation, and screenshot.
Contact	Contact type, interfaces, assumptions, and any nonlinear settings.
Mesh	Element type and order, global size, local sizes, growth control, node count, element count, and screenshot.
Equilibrium	Applied-load vector, support-reaction vector, imbalance, and percentage difference.
Stress	Maximum von Mises stress, location, stable nearby monitored stress, and contour screenshot.
Factor of safety	Minimum value, location, yield definition, and contour screenshot.
Displacement	Maximum magnitude, monitored displacement, direction, and location.
Convergence	At least three meshes, same monitored locations, result changes, and selected mesh justification.
Acceptance	FOS and displacement pass/fail, local failure-mode checks, limitations, and engineering disposition.

A.7 Bracket Corrected-Study Record Template

Table A.5: Bracket Static Stress input record

Field	Required entry
Material	Project Aluminum 6061-T6 properties
Analytical load	1471.5 N
Fusion force input	1472 N total
Force per entity	Off
Loaded faces	Four upper sloping rib faces
Load direction	Downward in the declared study coordinate system
Supports	Four mounting-hole cylindrical faces
Element order	Parabolic
Local refinements	Mounting holes and rib-root fillets
Reaction target	Approximately 1472 N opposite the applied direction
Constraint checks	FOS ≥ 2.5 and displacement ≤ 0.5 mm

A.8 Mesh-Convergence Template

Table A.6: Mesh-convergence recording template

Mesh	Nodes	Elements	Peak stress (MPa)	Monitored stress (MPa)	Displacement (mm)
1					
2					
3					
4					

Mesh 1 is the coarse reference, meshes 2 and 3 are successive refinements, and mesh 4 is an optional confirmation. The monitored stress must use the same geometric region on every mesh. A moving global maximum is not a valid convergence quantity when it moves between unrelated singular edges.

A.9 Stabilizer Pressure-Band Calculation

Let the unnormalized weights be

$$\mathbf{w} = [1.00, 0.93, 0.75, 0.40]. \quad (\text{A.1})$$

Their sum is 3.08. The band-force fractions are:

$$\hat{\mathbf{w}} = [0.324675, 0.301948, 0.243506, 0.129870]. \quad (\text{A.2})$$

For the 11 808 N resultant, the approximate band forces are:

Table A.7: Prescribed stabilizer pressure-band forces

Band	Normalized fraction	Force (N)
1	0.324675	3833.8
2	0.301948	3565.4
3	0.243506	2875.1
4	0.129870	1533.8
Total	1.000000	11 808.0

The FEA pressure for a band is its force divided by its selected CAD face area. The forces are prescribed engineering inputs and are not CFD results.

A.10 Evidence Inventory

Table A.8: Principal repository evidence used in the report

Location	Evidence
src/nemo/parts/definitions	Authoritative part parameters, units, materials, loads, constraints, fixed geometry, and boundary roles.
src/nemo/parts/analytical.py	First-order volume, stress, factor-of-safety, and displacement equations.
src/nemo/optimizer.py	Scaled bounded Nelder–Mead search and evaluation handling.
src/nemo/logger.py	Stable one-part-per-log CSV and JSON records.
fusion_addin/NEMOBridge	Fusion add-in, native generators, artifact export, and boundary selectors.
reports/final_20260728*_validation	Baseline and finalist parameters, analytical metrics, Fusion requests, Fusion responses, and checklists.
F360report/StudyFinal.html	Bracket manual study setup, mesh metadata, result views, and the recorded 1.472 N force error.
demos/bracket-proof-of-concept	Preserved initial demonstration flow and historical validation material.

Location	Evidence
reports/project-report	LaTeX report source, bibliography, extracted images, reproducible chart script, and compiled PDF.

A.11 Demonstration Sequence

A concise defense demonstration can follow this order:

1. state the problem and evidence boundary;
2. show the three part definitions;
3. evaluate the three analytical baselines;
4. show the 60-sample and three-start workflow;
5. compare baseline and finalist mass;
6. open one stored validation package;
7. send or replay a bracket CAD request;
8. show the native bracket, mass, STEP, and boundary metadata;
9. identify the four support faces and four upper load faces;
10. explain the 1.472 N error and the corrected 1472 N procedure;
11. show how padeye and stabilizer roles extend the same architecture;
12. close with the validation work required before manufacture.

A.12 Report Metadata Requiring Confirmation

The title-page project code remains MR-PR0-26-XX, and the supervisor is shown as SUPERVISOR NAME. These placeholders must be replaced with the official project code and supervisor name before final institutional submission.